# ADHM construction on hyper-Kähler manifolds of type $A_\infty$

Takeshi Yoneda
Department of Mathematics, Osaka University

**Abstract**

We obtain the ADHM construction of instantons on hyper-Kähler manifolds of type $A_\infty$, which is a generalization of the result of Kronheimer-Nakajima [11]. Our ADHM constuction reduces the problem of construction of instantons to the solution of a quadratic equation for certain linear operators of $L^2(S^1,\mathbb{C})$, which are associated with the Dynkin diagram of type $A_\infty$. We show that the $L^2$-norms of curvature of instantons on hyper-Kähler manifolds of type $A_\infty$ obtained by our ADHM construction are finite.

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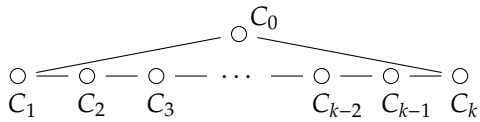
## 1 Introduction

The ADHM construction [2] is a well-known method of constructions of instantons, which enables us to construct complex vector bundles equipped with anti-self-dual connections from the solution of a quadratic equation for certain linear operators of $L^2(S^1,\mathbb{C})$. Kronheimer-Nakajima [11] obtained the ADHM construction on hyper-Kähler manifolds of type $A_k$, $D_k$, $E_6$, $E_7$ and $E_8$. Hyper-Kähler manifolds of type $A_\infty$ are constructed in [1] and [6], which are regarded as a certain limit of hyper-Kähler manifolds of type $A_k$ when $k$ goes infinity. In this paper we obtain the ADHM construction on hyper-Kähler manifolds of type $A_\infty$. Note that we restrict our attention to instantons whose structure groups are unitary as in [11].

Every hyper-Kähler manifold of type $A_k$ is diffeomorphic to the minimal resolution of the quotient singularity $\mathbb{C}^2/\Gamma_k$, where $\Gamma_k$ is the cyclic group of order $k+1$ in SU(2). The topological type of a hyper-Kähler manifold of type $A_k$ is expressed by the following diagram:

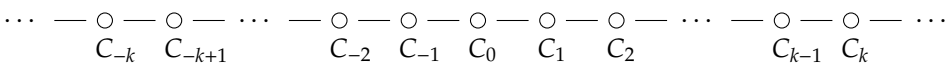
$$\begin{array}{ccccccccccc} \circ & — & \circ & — & \circ & — & \cdots & — & \circ & — & \circ & — & \circ \\ C_1 & & C_2 & & C_3 & & & & C_{k-2} & & C_{k-1} & & C_k \end{array}$$

where each circle denotes a rational curve $\mathbb{C}P^1$ whose self intersection number is $-2$, and two circles are joined by a line segment whenever the corresponding curves intersect. McKay observation also gives the extended Dynkin diagram of type $A_k$




from the regular representation $R_k$ of $\Gamma_k$. With this similarity between the topological type of hyper-Kähler manifolds of type $A_k$ and the regular representation of $\Gamma_k$, Kronheimer [10] constructed hyper-Kähler manifolds of type $A_k$ by the hyper-Kähler quotient method on a flat quaternion vector space $M = (Hom(R_k, R_k) \otimes \mathbb{H})^{\Gamma_k}$ with the action of $\mathrm{U}(R_k)$, where $\mathrm{U}(R_k)$ denotes the unitary group of $R_k$.

The topological type of a hyper-Kähler manifold of type $A_\infty$ is infinite, and expressed by the Dynkin diagram of type $A_\infty$:




where each circle denotes a rational curve $\mathbb{C}P^1$ whose self intersection number is $-2$, and two circles are joined by a line segment whenever the corresponding curves intersect. Goto [6] related this diagram to the regular representation $L^2(S^1, \mathbb{C})$ of $S^1$, which is the space of complex-valued $L^2$-functions on $S^1$, and constructed hyper-Kähler manifolds of type $A_\infty$ as infinite dimensional hyper-Kähler quotients on a certain infinite dimensional affine quaternion vector space with an action of a Hilbert Lie subgroup of $\mathrm{U}(L^2(S^1, \mathbb{C}))$, where $\mathrm{U}(L^2(S^1, \mathbb{C}))$ is the group of unitary operators on $L^2(S^1, \mathbb{C})$.

In the ADHM construction on hyper-Kähler manifolds of type $A_k$ [11], an ADHM data $(V, W, B, \Psi)$ consists of two finite dimensional $\Gamma_k$-modules $V$ and $W$, a suitable linear map $B$ on $V$, and a $\Gamma$-equivariant linear map $\Psi$ between $V$ and $W$. By McKay observation, the ADHM data can be identified with the set of the irreducible representations and the set of morphisms associated with the extended Dynkin diagram of type $A_k$. Our ADHM construction on hyper-Kähler manifolds of type $A_\infty$ in this paper also uses the similar definition of the ADHM data. Our ADHM data consists of two finite dimensional $S^1$-modules $V$ and $W$, a suitable linear operator $B$ on $V$, and a $S^1$-equivariant linear operator $\Psi$ between $V$ and $W$. Since the Dynkin diagram of type $A_\infty$ is infinite one, we can also consider ADHM data in which $V$ is an infinite dimensional $S^1$-module.

Our main results in this paper are summarized as follows. For the cases in which $V$ is a finite $S^1$-module, we show that an instanton on hyper-Kähler manifolds of type $A_\infty$ can be constructed from an ADHM data which satisfies certain algebraic equations (Theorem 3.5), and the $L^p$-norm of curvature of the instanton is finite for $p > \frac{6}{5}$ (Corollary 7.4). On the other hand, when $V$ is a certain infinite dimensional $S^1$-module, we show that an analytic sheaf whose each stalk is finite dimensional can be constructed from an ADHM data on hyper-Kähler manifolds of type $A_\infty$ (Corollary 4.8).

This paper is organized as follows. In section 2, we recall the construction of hyper-Kähler manifolds of type $A_\infty$ as hyper-Kähler quotients [6]. In section 3, we give our ADHM construction on hyper-Kähler manifolds of type $A_\infty$ for finite dimensional ADHM data in which $V$ is a finite dimensional $S^1$-modules. In section 4, we extend the result in section 3 to the case in which $V$ is a certain infinite dimensional $S^1$-module, and we show that finite dimensional holomorphic vector bundles are constructed from infinite dimensional ADHM data. We also obtain the result that analytic sheaves over $X_\zeta$ whose stalks have finite dimension. In section 5, we investigate the asymptotic behavior of the norms of curvature of instantons constructed by our ADHM construction. In section 6, a topological property of an instanton corresponding to a rational curve $\mathbb{C}P^1$ is proved. In addition, we get the useful description of hyper-Kähler symplectic forms which we make use of in the next section. In section 7, we verify the $L^p$-integrability of curvature obtained by finite dimensional ADHM data for $p > \frac{6}{5}$. In section 8, we provide some of examples of instantons and analytic sheaves constructed by our ADHM construction.

As a future work, we shall prove that the completeness of our ADHM construction on hyper-Kähler manifolds of type $A_\infty$ like the case of hyper-Kähler manifolds of type $A_k$ [11]. In other words, we shall prove that every instanton on hyper-Kähler manifolds of type $A_\infty$ is constructed by our ADHM

construction and the data is unique up to a certain equivalence relation on ADHM data. This problem seems much harder than the case with hyper-Kähler manifolds of type $A_k$ because the Riemannian metrics of hyper-Kähler manifolds of type $A_\infty$ are no longer asymptotically locally Euclidean, which is the property Kronheimer and Nakajima made use of in the proof of completeness with hyper-Kähler manifolds of type $A_k$ [11].

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## 2 Operator construction of hyper-Kähler manifolds of type $A_\infty$

In this section we review the operator construction of hyper-Kähler manifolds of type $A_\infty$, following [6].

Let $L^2(S^1,\mathbb{C}^m)$ be the Hilbert space of $\mathbb{C}^m$-valued $L^2$- functions on $S^1$, and we have the regular representation of $S^1$ on $L^2(S^1,\mathbb{C}^m)$ defined by

$$S^1 \times L^2(S^1,\mathbb{C}^m) \to L^2(S^1,\mathbb{C}^m)\ ,\ (g, f(\theta)) \mapsto f(g^{-1}\theta).$$

Here, we denote by $\theta$ the natural coordinate on $S^1 : e^{i\theta} \to \theta$. Then we can decompose $L^2(S^1,\mathbb{C}^m) = \oplus_{n\in\mathbb{Z}}\mathbb{C}^m \otimes V_n$, where $V_n$ is the one dimensional representation of $S^1$ generated by the function $e^{in\theta}$. Throughout this paper, we denote by $e_n$ the function $e^{in\theta}$, and us $\{e_n\}_{n\in\mathbb{Z}}$ as a complete orthogonal basis of $L^2(S^1,\mathbb{C}^m)$ for any $m \in \mathbb{N}$. We define the following vector spaces :

$$\begin{aligned}
Hom(L^2(S^1,\mathbb{C}^m)) &:= \left\{ a\ densely\ defined\ linear\ operator\ on\ L^2(S^1,\mathbb{C}^m) \right\} \\
B(L^2(S^1,\mathbb{C}^m)) &:= \left\{ a\ bounded\ linear\ operator\ on\ L^2(S^1,\mathbb{C}^m) \right\} \\
B^2(L^2(S^1,\mathbb{C}^m)) &:= \left\{ a\ Hilbert\ Schmidt\ linear\ operator\ on\ L^2(S^1,\mathbb{C}^m) \right\} \\
B_m(L^2(S^1,\mathbb{C}^m)) &:= \left\{ A \in B(L^2(S^1,\mathbb{C}^m)) \mid A(\mathbb{C}^m \otimes V_n) \subset \mathbb{C}^m \otimes V_{n+m},\ n \in \mathbb{Z} \right\} \\
B^2_m(L^2(S^1,\mathbb{C}^m)) &:= B_m(L^2(S^1,\mathbb{C}^m)) \cap B^2(L^2(S^1,\mathbb{C}^m))
\end{aligned}$$

Let us consider the Hilbert space $B^2(L^2(S^1,\mathbb{C}^m)) \otimes_{\mathbb{C}} Q$, where we define $Q = V_{-1} \oplus V_1$ and Q is naturally regarded as the quaternion. Clearly, we have the $S^1$ action on the Hilbert space.

**Lemma 2.1** (Lemma 6-1 in [6]). *Let $(B^2(L(S^1,\mathbb{C}^m)) \otimes_{\mathbb{C}} Q)^{S^1}$ be the Hilbert space of $S^1$-invariant elements of $B^2(L(S^1,\mathbb{C}^m)) \otimes_{\mathbb{C}} Q$. Then, we have*

$$(B^2(L^2(S^1,\mathbb{C}^m)) \otimes_{\mathbb{C}} Q)^{S^1} \quad = \quad (B^2_1(L^2(S^1,\mathbb{C}^m)) \otimes V_{-1}) \oplus (B^2_{-1}(L^2(S^1,\mathbb{C}^m)) \otimes V_1).$$

From now on, we focus on the Hilbert space $M := (B^2(L^2(S^1)) \otimes_{\mathbb{C}} Q)^{S^1}$, where $L^2(S^1)$ denotes $L^2(S^1,\mathbb{C})$ for simplicity. Since Q is the quaternion, the hyper-Kähler structure is naturally induced on $M$.

Let us define an densely defined operator $\Lambda = (\Lambda^+, \Lambda^-) \in (Hom(L^2(S^1)) \otimes_{\mathbb{C}} Q)^{S^1}$ by

$$\begin{aligned}
\Lambda^+ &= \phi \circ \frac{d}{d\theta} \circ p_+ \\
\Lambda^- &= p_- \circ \frac{d}{d\theta} \circ \psi
\end{aligned}$$

where $\psi^{-1} := \phi$ is defined by the multiplication with $e^{i\theta}$, and $p_\pm$ are defined by the orthogonal projections

$$\begin{aligned}
p_+ &: L^2(S^1) \to \left\{ f = \sum_{k\geq 0} a_k e_k \in L^2(S^1) \,\middle|\, a_k \in \mathbb{C}\ ,\ \sum_k |a_k|^2 < \infty \right\}, \\
p_- &: L^2(S^1) \to \left\{ f = \sum_{k<0} a_k e_k \in L^2(S^1) \,\middle|\, a_k \in \mathbb{C}\ ,\ \sum_k |a_k|^2 < \infty \right\}.
\end{aligned}$$

Now we have the affine hyper-Kähler Hilbert manifold modeled on $M$:

$$\hat{M} := \Lambda + M \ \subset\ (Hom(L^2(S^1)) \otimes_{\mathbb{C}} Q)^{S^1}.$$

In order to construct hyper-Kähler manifolds of type $A_\infty$ as hyper-Kähler quotients, we need to define a Hilbert Lie group $G$ which acts on $\hat{M}$ preserving its hyper-Kähler structure.
First, let $\mathfrak{u}(L^2(S^1))^{S^1}$ be the set of $S^1$-invariant skew symmetric operators on $L^2(S^1)$. For $X \in \mathfrak{u}(L^2(S^1))^{S^1}$, we denote by $X_n$ the eigen value of $X$ on $V_n$. Now we define the Hilbert Lie algebra $\mathfrak{g}$ by

$$\mathfrak{g} := \left\{ X \in \mathfrak{u}(L^2(S^1))^{S^1} \mid \lim_{|n|\to\infty} X_n = 0 \, , \, \left[ X, \phi \circ \tfrac{d}{d\theta} \right] \in B_1^2(L^2(S^1)) \right\}$$

whose inner product is given by

$$< X, Y >_{\mathfrak{g}} \quad = \quad \left\langle \left[ X, \phi \circ \frac{d}{d\theta} \right], \left[ Y, \phi \circ \frac{d}{d\theta} \right] \right\rangle_{B^2(L^2(S^1))} .$$

Finally we define the Hilbert Lie group G as the lie group generated by $\mathfrak{g}$. More precisely, $G$ is given by

$$G \quad := \quad \left\{ g \in \mathrm{U}(L^2(S^1))^{S^1} \mid \lim_{|n|\to\infty} g_n = 1 \, , \, \left[ g, \phi \circ \frac{d}{d\theta} \right] \in B_1^2(L^2(S^1)) \right\}$$

where we denote by $\mathrm{U}(L^2(S^1))^{S^1}$ the set of $S^1$-invariant unitary operators on $L^2(S^1)$ and by $g_n$ the eigen value of $g$ on $V_n$. The second condition in the definition of $G$ not only gives $G$ a Hilbert Lie group structure but also verifies the well-definedness of the following action:

$$G \times \hat{M} \to \hat{M} \, , \, (g, (\xi^+, \xi^-)) \mapsto (g \circ \xi^+ \circ g^{-1}, \, g \circ \xi^- \circ g^{-1}).$$

This action is obviously preserving the hyper-Kähler structure, and therefore there exists the unique hyper-Kähler moment map $\mu : \hat{M} \to \mathfrak{g}^* \otimes Im\mathbb{H}$ if we impose the condition $\mu(\Lambda) = 0$ since the space is affine. In this case, we can more explicitly write $\mu$ as $\mu = \mu_I i - \mu_{\mathbb{C}} k$ where $\mu_{\mathbb{R}}$ and $\mu_{\mathbb{C}}$ is given by

$$\begin{aligned} \mu_I(\xi^+, \xi^-) &= \frac{1}{2} \{ \, [\xi^+, (\xi^+)^*] + [\xi^-, (\xi^-)^*] - [\Lambda^+, (\Lambda^+)^*] - [\Lambda^-, (\Lambda^-)^*] \, \} \\ \mu_{\mathbb{C}}(\xi^+, \xi^-) &= [\xi^+, \xi^-] - [\Lambda^+, \Lambda^-]. \end{aligned}$$

Here we identify $\mathfrak{g}^*$ with $\mathfrak{g}$ by the inner product. Now we are ready to construct hyper-Kähler Manifolds of type $A_\infty$ as hyper-Kähler quotients on $(\hat{M}, G, \mu)$:

**Theorem 2.2** (Theorem 6-7 in [6])**.**
*Let $\zeta \in \mathfrak{g}^* \otimes Im\mathbb{H}$ be a generic element so that $G$ acts on $\mu^{-1}(\zeta)$ freely. Then the quotient space $X_\zeta := \mu^{-1}(\zeta)/G$ becomes a noncompact complete hyper-Kähler 4-manifold. Moreover, its topological type is given by the Dynkin diagram of type $A_\infty$, i.e., there exists a basis $\{C_n\}_{n\in\mathbb{Z}}$ of $H_2(X_\zeta, \mathbb{Z})$ such that the intersection form is given by*

$$< C_n, C_m > = \begin{cases} -2 & ( \, n = m \, ) \\ 1 & ( \, |n - m| = 1 \, ) \\ 0 & (otherwise) \end{cases}$$

In the following sections, we denote by $X_\zeta$ a hyper-ähler manifold of type $A_\infty$ given by the above theorem for $\zeta \in \mathfrak{g} \otimes Im\mathbb{H} \cong \mathfrak{g}^* \otimes Im\mathbb{H}$, and we decompose $\zeta$ as $\zeta = \zeta_{\mathbb{R}} i - \zeta_{\mathbb{C}} k \in \mathfrak{g} \otimes Im\mathbb{H}$ where $\zeta_{\mathbb{R}} \in \mathfrak{g}$ and $\zeta_{\mathbb{C}} \in \mathfrak{g} \otimes \mathbb{C}$.

# 3 Instantons from finite dimensional ADHM data

In this section, we obtain the ADHM construction on $X_\zeta$ associated with "finite" dimensional ADHM data, following the case with ALE spaces [11].

**Definition 3.1** (Finite dimensional ADHM data)**.**
*A data $(V,W,B,\Psi)$ is called a finite dimensional ADHM data if its elements satisfy the following conditions:*
*(1) $V$ and $W$ are finite dimensional $S^1$-modules*
*(2) $B=(B^+,B^-)\in(Hom(V,V)\otimes Q)^{S^1}$.*
*(3) $\Psi=(\Psi_1,\Psi_2)$, where $\Psi_1:W\to V$ and $\Psi_2:V\to W$ are $S^1$-equivariant linear maps.*
*(4) (Complex part of ADHM equation.)*

$$[B^+,B^-]+\Psi_1\Psi_2=-(\zeta_V)_{\mathbb{C}}.$$

*(5) (Real part of ADHM equation.)*

$$[B^+,(B^+)^*]+[B^-,(B^-)^*]+\Psi_1(\Psi_1)^*-(\Psi_2)^*\Psi_2=-(\zeta_V)_{\mathbb{R}}.$$

*Here the definition of $\zeta_V=(\zeta_V)_{\mathbb{R}}i+(\zeta_V)_{\mathbb{C}}j\in Hom_{S^1}(V,V)\otimes Im\mathbb{H}$ is as follows: For all $k\in\mathbb{Z}$, let us define $(\zeta_V)_k\in Im\mathbb{H}$ by the equation*

$$\{(\,[\Lambda^+,(\Lambda^+)^*]+[\Lambda^-,(\Lambda^-)^*]+2(\zeta_{\mathbb{R}})\,)i+(\,[\Lambda^+,\Lambda^-]+(\zeta_{\mathbb{C}})\,)j\}(e_{-k})=(\zeta_V)_k e_{-k}.$$

*Since $V$ is a $S^1$-module, we can decompose $V$ as $V=\oplus_k\mathbb{C}^{n_k}\otimes V_k$. Then we define $\zeta_V\in Hom(V,V)\otimes Im\mathbb{H}$ by*

$$\zeta_V:=(\zeta_V)_k id_{\mathbb{C}^{n_k}\otimes V_k}\qquad \text{on } \mathbb{C}^{n_k}\otimes V_k.$$

From now on, we fix a finite dimensional ADHM data $(V,W,B,\Psi)$ and construct a finite dimensional complex vector bundle equipped with a natural anti-self-dual connection.
First, let us consider $\mu^{-1}(\zeta)\to X_\zeta$ as a principal $G$-bundle, and we define $\mathcal{L}$ as the associated pre-Hilbert vector bundle given by the following action of $G$ on $C^\infty(S^1)$ with the $L^2$-metric:

$$G\times C^\infty(S^1)\longrightarrow C^\infty(S^1)\,,\ \Big(\,g=(g_n),\ \sum a_ne_n\Big)\longmapsto\sum a_ng_ne_n.$$

The metric is well-defined since the action is unitary. Furthermore, it inherits an anti-self-dual connection from the one on $\mu^{-1}(0)\to X_\zeta$ due to [5]. Then, we consider the following vector bundle complex on $X_\zeta$:

$$0\to\quad(V\otimes\mathcal{L})^{S^1}\xrightarrow{\sigma}(V\otimes\mathcal{L}\otimes Q)^{S^1}\oplus(W\otimes\mathcal{L})^{S^1}\xrightarrow{\tau}(V\otimes\mathcal{L})^{S^1}\to 0$$

where we denote by $(\ )^{S^1}$ the $S^1$-invariant part of vector bundles, and the definitions of $\sigma$ and $\tau$ are as follows: Let us define smooth maps $\alpha$ and $\beta$ by

$$\begin{aligned}\alpha\quad&:\mu^{-1}(\zeta)\to Hom_{S^1}(C^\infty(S^1),C^\infty(S^1)\otimes Q),\ (\xi^+,\xi^-)\longmapsto\Big(\,f\longmapsto\begin{pmatrix}\xi^+(f)\\ \xi^-(f)\end{pmatrix}\Big)\\ \beta\quad&:\mu^{-1}(\zeta)\to Hom_{S^1}(C^\infty(S^1)\otimes Q,C^\infty(S^1)),\ (\xi^+,\xi^-)\longmapsto\Big(\begin{pmatrix}f_1\\ f_2\end{pmatrix}\longmapsto\xi^-(f_1)-\xi^+(f_2)\,\Big).\end{aligned}$$

Since they are $G$-equivariant, so we can consider them as the smooth sections of $Hom_{S^1}(\mathcal{L},\mathcal{L}\otimes Q)$ and $Hom_{S^1}(\mathcal{L}\otimes Q,\mathcal{L})$ respectively. Then, we define $\sigma$ and $\tau$ by

$$\begin{aligned}\sigma\quad&:=\quad\left[\begin{pmatrix}B^+\\ B^-\end{pmatrix}\otimes id_{\mathcal{L}}-id_V\otimes\alpha\right]\oplus\Psi_2\otimes id_{\mathcal{L}}\\ \tau\quad&:=\quad[\,(B^-,-B^+)\otimes id_{\mathcal{L}}-id_V\otimes\beta\,]\oplus(-\Psi_1)\otimes id_{\mathcal{L}}.\end{aligned}$$

Note that the above sequence is actually complex, i.e., $\tau\sigma=0$ since the complex part of the ADHM equation is satisfied, and each vector bundle is finite dimensional. Now we consider the vector bundle morphism $\mathcal{D}$ defined by

$$\mathcal{D}\,:\,\mathbb{C}^2\otimes(V\otimes\mathcal{L})^{S^1}\ \to\ (V\otimes\mathcal{L}\otimes Q)^{S^1}\oplus(W\otimes\mathcal{L})^{S^1},\ (s,u)\mapsto\sigma(s)+\tau^*(u).$$

On a fibre at $p=[(\xi^+,\xi^-)]\in X_\zeta$, $\mathcal{D}$ is given by the following matrix:

$$\mathcal{D}_p=\left\{\begin{pmatrix} B^+ & (B^-)^* \\ B^- & -(B^+)^* \end{pmatrix}\otimes id_{\mathcal{L}}-id_V\otimes\begin{pmatrix} \xi^+ & (\xi^-)^* \\ \xi^- & -(\xi^+)^* \end{pmatrix}\right\}\oplus(\ \Psi_2,\ (-\Psi_1)^*).$$

**Lemma 3.2.** *There exists the vector bundle endmorphism $\Delta$ on $(V\otimes\mathcal{L})^{S^1}$ which satisfies the equation*

$$\mathcal{D}^*\mathcal{D}=id_{\mathbb{C}^2}\otimes\Delta.$$

*Proof.* For simplicity, we only prove the case where $\zeta=0$. Using the ADHM equations, we have

$$\begin{aligned}
(\mathcal{D}^*\mathcal{D})_p &= \begin{pmatrix} (B^+)^*B^++(B^-)^*B^- & -[B^+,B^-]^* \\ -[B^+,B^-] & B^+(B^+)^*+B^-(B^-)^* \end{pmatrix}\otimes id_{\mathcal{L}_p} \\
&\quad +id_V\otimes\begin{pmatrix} (\xi^+)^*\xi^++(\xi^-)^*\xi^- & -[\xi^+,\xi^-]^* \\ -[\xi^+,\xi^-] & \xi^+(\xi^+)^*+\xi^-(\xi^-)^* \end{pmatrix} \\
&\quad -\begin{pmatrix} (B^+)^*\otimes\xi^++(B^-)^*\otimes\xi^- & (B^+)^*\otimes(\xi^-)^*-(B^-)^*\otimes(\xi^+)^* \\ B^-\otimes\xi^+-B^+\otimes\xi^- & B^-\otimes(\xi^-)^*+B^+\otimes(\xi^+)^* \end{pmatrix} \\
&\quad -\begin{pmatrix} B^+\otimes(\xi^+)^*+B^-\otimes(\xi^-)^* & (B^-)^*\otimes(\xi^+)^*-(B^+)^*\otimes(\xi^-)^* \\ B^+\otimes\xi^--B^-\otimes\xi^+ & (B^-)^*\otimes\xi^-+(B^+)^*\otimes\xi^+ \end{pmatrix}+\begin{pmatrix} (\Psi_2)^*\Psi_2 & -(\Psi_1\Psi_2)^* \\ -\Psi_1\Psi_2 & \Psi_1(\Psi_1)^* \end{pmatrix} \\
&= \begin{pmatrix} (B^+)^*B^++(B^-)^*B^- & 0 \\ 0 & B^+(B^+)^*+B^-(B^-)^* \end{pmatrix}\otimes id_{\mathcal{L}_p} \\
&\quad +id_V\otimes\begin{pmatrix} (\xi^+)^*\xi^++(\xi^-)^*\xi^- & 0 \\ 0 & \xi^+(\xi^+)^*+\xi^-(\xi^-)^* \end{pmatrix} \\
&\quad -\begin{pmatrix} (B^+)^*\otimes\xi^++(B^-)^*\otimes\xi^- & 0 \\ 0 & B^-\otimes(\xi^-)^*+B^+\otimes(\xi^+)^* \end{pmatrix} \\
&\quad -\begin{pmatrix} B^+\otimes(\xi^+)^*+B^-\otimes(\xi^-)^* & 0 \\ 0 & (B^-)^*\otimes\xi^-+(B^+)^*\otimes\xi^+ \end{pmatrix}+\begin{pmatrix} (\Psi_2)^*\Psi_2 & 0 \\ 0 & \Psi_1(\Psi_1)^* \end{pmatrix}.
\end{aligned}$$

Then if we put $(\mathcal{D}^*\mathcal{D})_p=\begin{pmatrix} \tilde{\Delta}_p & 0 \\ 0 & \Delta_p \end{pmatrix}$, the difference $\Delta_p-\tilde{\Delta}_p$ is calculated as:

$$\begin{aligned}
\Delta_p-\tilde{\Delta}_p &= ([B^+,(B^+)^*]+[B^-,(B^-)^*]+\Psi_1(\Psi_1)^*-(\Psi_2)^*\Psi_2)\otimes id_{\mathcal{L}_p}+id_V\otimes([\xi^+,(\xi^+)^*]+[\xi^-,(\xi^-)^*]) \\
&= -(\zeta_V)_{\mathbb{R}}\otimes id_{\mathcal{L}_p}+id_V\otimes([\Lambda^+,(\Lambda^+)^*]+[\Lambda^-,(\Lambda^-)^*]) \\
&= 0.
\end{aligned}$$

□

**Remark 3.3.** *We denote by $I,J,K$ the hyper-Kähler complex structures on $X_\zeta$ and we also use the same notation for the canonical hyper-Kähler complex structures on $\mathbb{C}^2$ given by*

$$I=\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix},\qquad J=\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},\qquad K=\begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}.$$

*Note that $\mathcal{D}^*\mathcal{D}=id_{\mathbb{C}^2}\otimes\Delta$ and any of these complex structures on $\mathbb{C}^2$ are commutable.*

**Lemma 3.4.** *For any finite dimensional ADHM data, $\mathcal{D}$ is everywhere injective.*

*Proof.* Suppose its injectivity fails at a point, say, $[\xi=(\xi^+,\xi^-)]\in X_\zeta$. Then, there exists a non zero element $(\gamma_1,\gamma_2)\in\mathbb{C}^2\otimes(V\otimes L^2(S^1))^{S^1}$ such that

$$\sigma_\xi(\gamma_1)\ =\ \tau_\xi^*(\gamma_2)\ =\ 0$$

where $\sigma_\xi$ and $\tau_\xi$ are defined as the trivialization of $\sigma$ and $\tau$ on $[\xi]$. Moreover, since $\mathcal{D}_\xi = id_{\mathbb{C}^2} \otimes \sigma_\xi^* \sigma_\xi = id_{\mathbb{C}^2} \otimes \tau_\xi \tau_\xi^*$ due to lemma 3.2., it follows that

$$\sigma_\xi(\gamma_2) \;=\; \tau_\xi^*(\gamma_1) \;=\; 0.$$

On the other hand, let us regard $\gamma_1, \gamma_2$ as elements in $B(L^2(S^1), V)$, the space of bounded linear operators from $L^2(S^1)$ to $V$, by the following map:

$$(V \otimes L^2(S^1))^{S^1} \;\longrightarrow\; B(L^2(S^1), V), \;\; \sum_{i \in S_V} \sum_{j=1}^{m_i} v_i^j \otimes f_i^j \;\longmapsto \left( f \mapsto < f, f_i^j > v_i^j \right)$$

where we decompose $V$ as $V = \oplus_{i \in S_V} \mathbb{C}^{m_i} \otimes V_i$ and $\{v_i^j\}$ is a basis of $\mathbb{C}^{m_i} \otimes V_i$. In the same way, we also consider $(V \otimes L^2(S^1) \otimes Q)^{S^1}$ as a subset in $B(L^2(S^1), V) \otimes Q$. Then we have, for $\gamma = \gamma_1$ or $\gamma_2$,

$$\begin{pmatrix} B^+ \circ \gamma \\ B^- \circ \gamma \end{pmatrix} - \begin{pmatrix} \gamma \circ (\xi^+)^* \\ \gamma \circ (\xi^-)^* \end{pmatrix} = 0\,, \;\; \begin{pmatrix} (B^-)^* \circ \gamma \\ -(B^+)^* \circ \gamma \end{pmatrix} - \begin{pmatrix} \gamma \circ \xi^- \\ \gamma \circ (-\xi^+) \end{pmatrix} = 0,$$

as elements in $B(L^2(S^1), V) \otimes Q$. From the above equations, we see that $Ker\gamma_1, Ker\gamma_2 \subset L^2(S^1)$ are preserved by $\xi^+, (\xi^+)^*, \xi^-, (\xi^-)^*$. Furthermore, it easily follows that there exist finite sets $S_i \subset \mathbb{Z}$(i=1,2) such that $Ker\gamma_i = \oplus_{k \notin S_i} V_k$, and at least one of $S_1, S_2$ is not empty since we assume $(\gamma_1, \gamma_2) \neq 0$. Thus we can construct an non trivial element in $G$ which preserves $\xi \in \mu^{-1}(\zeta)$. This is a contradiction to the fact $G$ freely acts on $\mu^{-1}(\zeta)$. □

**Theorem 3.5.** *Let $(V, W, B, \Psi)$ be a finite dimensional ADHM data. Then $E := Coker\mathcal{D} \cong Ker\mathcal{D}^* \cong Ker\tau / Im\sigma$ has a complex vector bundle structure whose dimension equals* $\dim W$ *and its induced connection is anti-self-dual.*

*Proof.* The proof is exactly the same as the case with ALE spaces [11]. To begin with, $E$ is well-defined as a smooth vector bundle because of lemma 3.4, and it is clear from the construction that $\dim E = \dim W$. In order to prove the self duality of the induced connection, we start by giving the following notation:

$$\begin{aligned}
\pi_E &: (V \otimes \mathcal{L} \otimes Q)^{S^1} \oplus (W \otimes \mathcal{L})^{S^1} \to E \;\; : \;\; \textit{orthogonal projection}. \\
\pi_Q &: (V \otimes \mathcal{L} \otimes Q)^{S^1} \oplus (W \otimes \mathcal{L})^{S^1} \to (V \otimes \mathcal{L} \otimes Q)^{S^1} \;\; : \;\; \textit{orthogonal projection}. \\
\nabla &: \textit{the natural connection on } (\mathbb{C}^2 \otimes V \otimes \mathcal{L})^{S^1} \textit{ and } (V \otimes \mathcal{L} \otimes Q)^{S^1} \oplus (W \otimes \mathcal{L})^{S^1}. \\
\nabla^E &: \textit{the induced connection on } E \\
R,\, R^E &: \textit{the curvature of } \nabla \textit{ and } \nabla^E, \textit{ respectively}. \\
H &:= (\mathcal{D}^* \mathcal{D})^{-1} = \; id_V \otimes \Delta^{-1}. \\
id_V \otimes \xi &:= id_V \otimes (\alpha, \beta^*) \in \Gamma\left( Hom(\mathbb{C}^2 \otimes (V \otimes \mathcal{L})^{S^1}, (V \otimes \mathcal{L} \otimes Q)^{S^1}) \right).
\end{aligned}$$

Note that $\nabla$ is canonically anti-self-dual, i.e., it satisfies the equation

$$R(v, w) = R(Iv, Iw) = R(Jv, Jw) = R(Kv, Kw) \quad \text{for any } v, w \in \Gamma(TX_\zeta).$$

Now we have $\pi_E = 1 - \mathcal{D} \circ F \circ \mathcal{D}^*$ and the second fundamental form $\Pi$ can be written as

$$\begin{aligned}
\Pi \;&=\; (1 - \pi_E) \circ \nabla \\
&=\; \mathcal{D} \circ H \circ (\nabla(id_V \otimes \xi))^* \circ \pi_Q,
\end{aligned}$$

and it follows from the matrix form of $\mathcal{D}$ that

$$\begin{aligned}
\nabla_{Iv}(id_V \otimes \xi) \;&=\; \nabla_v(id_V \otimes \xi) \circ I \otimes id_{(V \otimes \mathcal{L})^{S^1}}, \\
\nabla_{Jv}(id_V \otimes \xi) \;&=\; \nabla_v(id_V \otimes \xi) \circ J \otimes id_{(V \otimes \mathcal{L})^{S^1}}, \\
\nabla_{Kv}(id_V \otimes \xi) \;&=\; \nabla_v(id_V \otimes \xi) \circ (-K) \otimes id_{(V \otimes \mathcal{L})^{S^1}}.
\end{aligned}$$

Thus, for any $\mathrm{A} \in \{I, J, K\}$, we have

$$\begin{aligned}
\Pi(Av)^* \circ \Pi(Aw) &= \pi_Q^* \circ \nabla_{Av}(id_V \otimes \xi) \circ H \circ (\nabla_{Aw}(id_V \otimes \xi))^* \circ \pi_Q \\
&= \pi_Q^* \circ \nabla_v(id_V \otimes \xi) \circ (A \otimes id_{(V\otimes\mathcal{L})^{S^1}}) \circ H \circ (A \otimes id_{(V\otimes\mathcal{L})^{S^1}})^* \circ (\nabla_w(id_V \otimes \xi))^* \circ \pi_Q \\
&= \pi_Q^* \circ \nabla_v(id_V \otimes \xi) \circ H \circ (A \otimes id_{(V\otimes\mathcal{L})^{S^1}}) \circ (-A \otimes id_{(V\otimes\mathcal{L})^{S^1}}) \circ (\nabla_w(id_V \otimes \xi))^* \circ \pi_Q \\
&= \pi_Q^* \circ \nabla_v(id_V \otimes \xi) \circ H \circ (\nabla_w(id_V \otimes \xi))^* \circ \pi_Q \\
&= \Pi(v)^* \circ \Pi(w).
\end{aligned}$$

On the other hand, for any sections $s, u \in \Gamma(E)$ and $v, w \in \Gamma(TX_\zeta)$ we have

$$(R^E(v,w)s, u) = (R(v,w)s, u) - (\Pi(v)s, \Pi(w)u) + (\Pi(w)s, \Pi(v)u)$$

due to the Gauss equation. Therefore we conclude that

$$R^E(v,w) = R^E(Iv, Iw) = R^E(Jv, Jw) = R^E(Kv, Kw)\,, \quad \text{for any } v, w \in \Gamma(TX_\zeta).$$

□

# 4 Analytic sheaves from infinite dimensional ADHM data

In this section, we construct a Hilbert vector bundle complex over $X_\zeta$ from an "infinite" dimensional ADHM data and we prove that its Euler characteristic is finite. After that, we see that the middle cohomology of the complex inherits a finite dimensional vector bundle structure under some condition, and its induced connection is compatible with $I$ on $X_\zeta$. More generally, we construct an analytic sheaf on $(X_\zeta, I)$from an infinite dimensional ADHM data and prove that its stalks are finite dimensional. The reason why we do not (or can not) construct instatons but holomorphic vector bundles is mentioned in Remark 4.7. For the sake of simplicity, we only consider the case where $\mu(\Lambda) = \zeta$ in the this section.

Let us define a densely defined linear operator $\tilde{\Lambda} = (\tilde{\Lambda}^+, \tilde{\Lambda}^-) \in (Hom(L^2(S^1, \mathbb{C}^m)) \otimes_{\mathbb{C}} Q)^{S^1}$ by

$$\begin{aligned} \tilde{\Lambda}^+ &= \phi \circ \frac{d}{d\theta} \circ p_+ \,, \\ \tilde{\Lambda}^- &= p_- \circ \frac{d}{d\theta} \circ \psi \,. \end{aligned}$$

This is a natural extension of $\Lambda \in (Hom(L^2(S^1)) \otimes_{\mathbb{C}} Q)^{S^1}$ which is used in the construction of $X_\zeta$.

**Definition 4.1** (Infinite dimensional ADHM data)**.**
*A data $(V, W, B, \Psi)$ is called an infinite dimensional ADHM data if its elements satisfy the following conditions:*
*(1) $V = L^2(S^1, \mathbb{C}^m)$, and $W$ is a finite dimensional $S^1$-module*
*(2) $B = (B^+, B^-) \in \tilde{\Lambda} + (B^2(L^2(S^1, \mathbb{C}^m)) \otimes Q)^{S^1} \subset Hom(L^2(S^1, \mathbb{C}^m)) \otimes Q$*
*(3) $\Psi = (\Psi_1, \Psi_2)$, where $\Psi_1 : W \to V$ and $\Psi_2 : V \to W$ are $S^1$- equivariant linear maps.*
*(4) (Complex part of ADHM equation.)*

$$[B^+, B^-] + \Psi_1 \Psi_2 = -(\zeta_v)_{\mathbb{C}} \ ( = 0 ).$$

Given an infinite dimensional ADHM data $(V, W, B, \Psi)$, we can construct a Hilbert vector bundle complex over $X_\zeta$. First of all, we consider the following pre-Hilbert vector bundle complex over $X_\zeta$:

$$0 \to \quad (C^\infty(S^1, \mathbb{C}^m) \otimes \mathcal{L})^{S^1} \xrightarrow{\sigma} \begin{array}{c} (C^\infty(S^1, \mathbb{C}^m) \otimes \mathcal{L} \otimes Q)^{S^1} \\ \bigoplus \\ (W \otimes \mathcal{L})^{S^1} \end{array} \xrightarrow{\tau} (C^\infty(S^1, \mathbb{C}^m) \otimes \mathcal{L})^{S^1} \to 0.$$

The definitions of $\sigma$ and $\tau$ are exactly the same as in the previous section, i.e., we define $\sigma$ and $\tau$ by

$$\begin{aligned} \sigma &:= \left[ \begin{pmatrix} B^+ \\ B^- \end{pmatrix} \otimes id_{\mathcal{L}} - id_{C^\infty(S^1, \mathbb{C}^m)} \otimes \alpha \right] \oplus \Psi_2 \otimes id_{\mathcal{L}} \\ \tau &:= [\ (B^-, -B^+) \otimes id_{\mathcal{L}} - id_{C^\infty(S^1, \mathbb{C}^m)} \otimes \beta\ ] \oplus (-\Psi_1) \otimes id_{\mathcal{L}} \end{aligned}$$

Note that the above sequence is actually complex, i.e., $\tau\sigma = 0$ if and only if the complex part of the ADHM equation is satisfied. Then, let us take the "$L^2$- completion" of the above pre-Hilbert vector bundle complex $\mathcal{E}$ :

$$\mathcal{E} \ : \ 0 \to \quad \overline{(C^\infty(S^1, \mathbb{C}^m) \otimes \mathcal{L})^{S^1}} \xrightarrow{\sigma} \begin{array}{c} \overline{(C^\infty(S^1, \mathbb{C}^m) \otimes \mathcal{L} \otimes Q)^{S^1}} \\ \bigoplus \\ (W \otimes \mathcal{L})^{S^1} \end{array} \xrightarrow{\tau} \overline{(C^\infty(S^1, \mathbb{C}^m) \otimes \mathcal{L})^{S^1}} \to 0$$

where, $\tau$ and $\sigma$ are regarded as its suitable closed extension. In the following, we prove that this complex has finite "Euler characteristic" for any choice of $(B^+, B^-, \Psi)$.
To clarify the mean of "$L^2$-completion", we need to trivialize the vector bundle complex. For each $\xi \in \mu^{-1}(0)$, we define a Hilbert complex $\mathcal{E}(B, \xi, \Psi)$ as the fibre of $\mathcal{E}$ on $[\xi]$:

$$0 \to \overline{(C^\infty(S^1, \mathbb{C}^m) \otimes C^\infty(S^1))^{S^1}} \xrightarrow{\sigma(B,\xi,\Psi)} \begin{array}{c} \overline{(C^\infty(S^1, \mathbb{C}^m) \otimes C^\infty(S^1) \otimes Q)^{S^1}} \\ \bigoplus \\ (W \otimes V)^{S^1} \end{array} \xrightarrow{\tau(B,\xi,\Psi)} \overline{(C^\infty(S^1, \mathbb{C}^m) \otimes C^\infty(S^1))^{S^1}} \to 0$$

We shall explicitly describe these Hilbert spaces and morphisms which appear in $\mathcal{E}(B,\xi,\Psi)$. With the natural complete orthogonal bases of $L^2(S^1)$ and $L^2(S^1,\mathbb{C}^m)$, each Hilbert space is be defined as :

$$\begin{aligned}
\overline{(C^\infty(S^1,\mathbb{C}^m)\otimes C^\infty(S^1))^{S^1}} &= \left\{ \sum_{k\in\mathbb{Z}} a_k e_k \otimes e_{-k} \;\middle|\; a_k \in \mathbb{C}^m \,, \sum_{k\in\mathbb{Z}} |a_k|^2 < \infty \right\} \\
&\cong L^2(S^1,\mathbb{C}^m) = V \\
\overline{(C^\infty(S^1,\mathbb{C}^m)\otimes C^\infty(S^1)\otimes Q)^{S^1}} &= \left\{ \sum_{k\in\mathbb{Z}} a_k e_k \otimes e_{-k+1} \;\middle|\; a_k \in \mathbb{C}^m \,, \sum_{k\in\mathbb{Z}} |a_k|^2 < \infty \right\} \\
&\quad \oplus \left\{ \sum_{k\in\mathbb{Z}} a_k e_k \otimes e_{-k-1} \;\middle|\; a_k \in \mathbb{C}^m \,, \sum_{k\in\mathbb{Z}} |a_k|^2 < \infty \right\} \\
&\cong L^2(S^1,\mathbb{C}^m) \oplus L^2(S^1,\mathbb{C}^m) = V \oplus V
\end{aligned}$$

where the isomorphisms are given by

$$\begin{aligned}
\left\{ \textstyle\sum_{k\in\mathbb{Z}} a_k e_k \otimes e_{-k} \;\middle|\; a_k \in \mathbb{C}^m \,, \sum_{k\in\mathbb{Z}} |a_k|^2 < \infty \right\} &\xrightarrow{\sim} V, \;\; e_k \otimes e_{-k} \longmapsto e_k, \\
\left\{ \textstyle\sum_{k\in\mathbb{Z}} a_k e_k \otimes e_{-k+1} \;\middle|\; a_k \in \mathbb{C}^m \,, \sum_{k\in\mathbb{Z}} |a_k|^2 < \infty \right\} &\xrightarrow{\sim} V, \;\; e_k \otimes e_{-k+1} \longmapsto e_k, \\
\left\{ \textstyle\sum_{k\in\mathbb{Z}} a_k e_k \otimes e_{-k-1} \;\middle|\; a_k \in \mathbb{C}^m \,, \sum_{k\in\mathbb{Z}} |a_k|^2 < \infty \right\} &\xrightarrow{\sim} V, \;\; e_k \otimes e_{-k-1} \longmapsto e_k.
\end{aligned}$$

On these identifications, $\sigma(B,\xi,\Psi)$ and $\tau(B,\xi,\Psi)$ are written as

$$\begin{aligned}
\sigma(B,\xi,\Psi) &= \begin{pmatrix} B^+ - \phi\circ\nu\circ\xi^+\circ\nu \\ B^- - \psi\circ\nu\circ\xi^-\circ\nu \\ \Psi_2 \end{pmatrix} : V \to V \oplus V \oplus (W\otimes C^\infty(S^1))^{S^1}, \\
\tau(B,\xi,\Psi) &= (B^- - \nu\circ\xi^-\circ\phi\circ\nu,\; -B^+ + \nu\circ\xi^+\circ\psi\circ\nu,\; -\Psi_1) \;:\; V\oplus V\oplus (W\otimes C^\infty(S^1))^{S^1} \to V,
\end{aligned}$$

where $\nu : V \to V$ is defined by $\nu(e_n) = e_{-n}$ and $\xi^\pm$ are naturally considered as elements in $Hom(V)$. Note that the domains of $\sigma(B,\xi,\Psi)$ and $\tau(B,\xi,\Psi)$ are defined as $L^2_2(S^1,\mathbb{C}^m)$ and $L^2_1(S^1,\mathbb{C}^m)\oplus L^2_1(S^1,\mathbb{C}^m)\oplus(W\otimes C^\infty(S^1))^{S^1}$.

For convenience, we shall give more explicit expressions of these morphisms: Let us define $(B^\pm)_n \in Hom(\mathbb{C}^m,\mathbb{C}^m)$ and $(\xi^\pm)_n \in \mathbb{C}$ by

$$\begin{aligned}
B^+(a_n e_n) &= (B^+)_n(a_n)e_{n+1}\,, & \xi^+(e_n) &= (\xi^+)_n e_{n+1} \\
B^-(a_{n+1}e_{n+1}) &= (B^-)_n(a_{n+1})e_n, & \xi^-(e_{n+1}) &= (\xi^-)_n e_n
\end{aligned}$$

where $a_n e_n \in \mathbb{C}^m \otimes V_n$. Then, for $p = \sum_{k\in\mathbb{Z}} a_n e_n \in Dom(\sigma(B,\xi,\Psi))$ or $Dom(\sigma(B,\xi,\Psi)^*)$, we have

$$\begin{aligned}
\sigma(B,\xi,\Psi)(p) &= \begin{pmatrix} \sum_{n\in\mathbb{Z}} [(B^+)_{n-1}(a_{n-1}) - (\xi^+)_{-n}\, a_n] e_n \\ \sum_{n\in\mathbb{Z}} [(B^-)_n(a_{n+1}) - (\xi^-)_{-n-1}\, a_n] e_n \\ -\Psi_2(p) \end{pmatrix}, \\
\tau(B,\xi,\Psi)^*(p) &= \begin{pmatrix} \sum_{n\in\mathbb{Z}} \Big[(B^-)^*_{n-1}(a_{n-1}) - (\xi^-)^*_{-n}\, a_n\Big] e_n \\ \sum_{n\in\mathbb{Z}} \Big[-(B^+)^*_n(a_{n+1}) + (\xi^+)^*_{-n-1}\, a_n\Big] e_n \\ \Psi_1^*(p) \end{pmatrix}.
\end{aligned}$$

**Theorem 4.2.** *For any infinite dimensional ADHM data and $\xi \in \mu^{-1}(0)$, $\mathcal{E}(B,\xi,\Psi)$ is Fredholm complex and its Euler characteristic $\chi(\mathcal{E}(B,\xi,\Psi))$ is* $\dim W$. *In other words, the followings are satisfied for any infinite dimensional ADHM data and $\xi \in \mu^{-1}(0)$.*
*(1) $\tau(B,\xi,\Psi)$ and $\sigma(B,\xi,\Psi)$ have closed ranges.*
*(2) Each cohomology $H_i(\mathcal{E}(B,\xi,\Psi))$ (i=0,1,2) of the complex $\mathcal{E}(B,\xi,\Psi)$ has finite dimension.*
*(3) $\chi(\mathcal{E}(B,\xi,\Psi)) := \sum_{i=0}^2 (-1)^{i+1}\dim H_i(\mathcal{E}(B,\xi,\Psi)) = \dim W$.*

**Remark 4.3.** *If (1) holds,*

$$\begin{aligned} Ker(\tau(B,\xi,\Psi))/Im(\sigma(B,\xi,\Psi)) &\cong Ker(\tau(B,\xi,\Psi))\cap Ker(\sigma(B,\xi,\Psi)^*) \\ \overline{(C^\infty(S^1,\mathbb{C}^m)\otimes C^\infty(S^1))^{S^1}}/Im(\tau(B,\xi,\Psi)) &\cong Ker(\tau(B,\xi,\Psi)^*). \end{aligned}$$

**Remark 4.4.** *Acoording to [3],*

$$(1) \text{ and } (2) \text{ hold} \quad \Leftrightarrow \quad \mathcal{D}(B,\xi,\Psi) := (\sigma(B,\xi,\Psi),\tau(B,\xi,\Psi)^*) \text{ is Fredholm operator,}$$

*and in this case, we have index*$(\mathcal{D}(B,\xi,\Psi)) = \chi(\mathcal{E}(B,\xi,\Psi))$.

In order to prove the theorem, we see that it results from the following proposition :
**Proposition 4.5.** *Let us consider the infinite dimensional ADHM data* $(V,W,B_0,(0,0))$ *and the element* $\xi_0 \in \mu^{-1}(0)$ *defined by*

$$\begin{aligned} B_0^+ &:= \tilde{\Lambda}^+ + p_0 + \left(\frac{d}{d\theta}\right)^{-1}\circ p_- , \qquad & \xi_0^+ &:= \Lambda^+ + p_0 \\ B_0^- &:= \tilde{\Lambda}^- + p_0 + \left(\frac{d}{d\theta}\right)^{-1}\circ p_{\geq 1} , \qquad & \xi_0^- &:= \Lambda^- \end{aligned}$$

*where* $p_{\geq 1}$ *is defined by the projection to* $\{\sum_{k\geq 1} a_k e_k \mid a_k \in \mathbb{C}^m \ , \ \Sigma|a_k|^2 < \infty\}$, *and* $p_0$ *is the projection to* $\mathbb{C}^m \otimes V_0$ *or* $V_0$. *Then, (1) (2) (3) in Theorem 4.2 are satisfied for* $(V,W,B_0,(0,0))$ *and* $\xi_0$.

Under the assumption that Proposition 4.5 holds, we can easily prove that Theorem 4.2 holds: For any infinite dimensional ADHM data $(V,W,B,\Psi)$ and $\xi \in \mu^{-1}(\zeta)$, $B$ and $\xi$ can be written as $B_0 +$ (*Hilbert-Schmidt Operator*) and $\xi_0 +$ (*Hilbert-Schmidt Operator*). Then, it follows that $\mathcal{D}(B,\xi,\Psi) = \mathcal{D}(B_0,\xi_0,0) +$ (*Compact Operator*) and therefore we can obtain the theorem because of Remark 4.4 and the fact that the sum of a densely defined Fredholm operator and a bounded compact operator is again Fredholm operator [12].

*Proof of Proposition 4.5.*
(1) Closedness of the ranges
By the definitions, $\tau(B_0,\xi_0,0)$ and $\sigma(B_0,\xi_0,0)$ are written as

$$\begin{aligned} \sigma(B_0,\xi_0,0) &= \begin{pmatrix} B_0^+ - \phi\circ\nu\circ\xi_0^+\circ\nu \\ B_0^- - \psi\circ\nu\circ\xi_0^-\circ\nu \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} \tilde{\Lambda}^+ - \phi\circ\nu\circ\Lambda^+\circ\nu \\ \tilde{\Lambda}^- - \psi\circ\nu\circ\Lambda^-\circ\nu \\ 0 \end{pmatrix} + \begin{pmatrix} \textit{Hilbert-Schmidt Operator} \\ \textit{Hilbert-Schmidt Operator} \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} \phi\circ\frac{d}{d\theta}\circ p_+ + \frac{d}{d\theta}\circ p_- \\ p_-\circ\frac{d}{d\theta}\circ\psi - \psi\circ\frac{d}{d\theta}\circ\phi\circ p_+ \\ 0 \end{pmatrix} + \begin{pmatrix} \textit{Hilbert-Schmidt Operator} \\ \textit{Hilbert-Schmidt Operator} \\ 0 \end{pmatrix} \end{aligned}$$

and

$$\begin{aligned} \tau(B_0,\xi_0,0) &= (B_0^- - \nu\circ\xi_0^-\circ\phi\circ\nu,\ -B_0^+ + \nu\circ\xi_0^+\circ\psi\circ\nu,\ 0) \\ &= (\tilde{\Lambda}^- - \nu\circ\Lambda^-\circ\phi\circ\nu,\ -\tilde{\Lambda}^+ + \nu\circ\Lambda^+\circ\psi\circ\nu,\ 0) \\ &\quad +(\textit{Hilbert-Schmidt Operator},\ \textit{Hilbert-Schmidt Operator},\ 0) \\ &= \left(p_-\circ\frac{d}{d\theta}\circ\psi + p_+\circ\frac{d}{d\theta},\ -\phi\circ\frac{d}{d\theta}\circ p_+ - \psi\circ\frac{d}{d\theta}\circ\phi\circ p_-,\ 0\right) \\ &\quad +(\textit{Hilbert-Schmidt Operator},\ \textit{Hilbert-Schmidt Operator},\ 0). \end{aligned}$$

It follows from a simple calculation that each components of the first terms is Fredholm operator. Thus, we conclude that all of components in $\sigma(B_0,\xi_0,0)$ and $\tau(B_0,\xi_0,0)$ are Fredholm operator due to the

fact that the sum of a densely defined Fredholm operator and a bounded compact operator is again Fredholm[12]. Therefore it follows that they have closed ranges.

(2),(3) Finiteness of cohomologies and Euler characteristic.
By the definitions, we have

$$(B_0^+)_n = \begin{cases} in\ id_{\mathbb{C}^m} & (\ n \geqq 1\ ) \\ id_{\mathbb{C}^m} & (\ n = 0\ ) \\ \frac{1}{in}\ id_{\mathbb{C}^m} & (\ n \leqq -1\ ) \end{cases} \qquad (\xi_0^+)_n = \begin{cases} in & (\ n \geqq 1\ ) \\ 1 & (\ n = 0\ ) \\ 0 & (\ n \leqq -1\ ) \end{cases}$$

$$(B_0^-)_n = \begin{cases} \frac{1}{in}\ id_{\mathbb{C}^m} & (\ n \geqq 1\ ) \\ id_{\mathbb{C}^m} & (\ n = 0\ ) \\ in\ id_{\mathbb{C}^m} & (\ n \leqq -1\ ) \end{cases} \qquad (\xi_0^-)_n = \begin{cases} 0 & (\ n \geqq 0\ ) \\ in & (\ n \leqq -1\ ). \end{cases}$$

(From this expression, it is clear that the data satisfies the complex part of the ADHM equation.)
First, we prove the injectivity of $\sigma(B_0, \xi_0, 0)$. On the above identification, we have

$$\begin{aligned} \sigma(B_0,\xi_0,0)(p) = 0 \quad &\Leftrightarrow \quad \begin{cases} (B_0^+)_{n-1}(a_{n-1}) = (\xi_0^+)_{-n}(a_n) \\ (B_0^-)_n(a_{n+1}) = (\xi_0^-)_{-n-1}(a_n) \end{cases} \quad \forall n \in \mathbb{Z} \\ &\Rightarrow \quad \begin{cases} in\ a_{n-1} = 0 & (\ n \geq 1\ ) \\ in\ a_{n+1} = 0 & (\ n \leq -2\ ) \end{cases} \\ &\Rightarrow \quad a_n = 0 \quad \forall n \in \mathbb{Z} \end{aligned}$$

Thus we have $\dim H_0(\mathcal{E}(B_0,\xi_0,0)) = \dim Ker(\sigma(B_0,\xi_0,0)) = 0$. In the same way, we have

$$\begin{aligned} \tau(B_0,\xi_0,0)^*(p) = 0 \quad &\Leftrightarrow \quad \begin{cases} (B_0^-)^*_{n-1}(a_{n-1}) = (\xi_0^-)^*_{-n}(a_n) \\ (B_0^+)^*_n(a_{n+1}) = (\xi_0^+)^*_{-n-1}(a_n) \end{cases} \quad \forall n \in \mathbb{Z} \\ &\Rightarrow \quad \begin{cases} in\ a_{n-1} = 0 & (\ n \leq 0\ ) \\ a_0 = a_{-1} \\ in\ a_{n+1} = 0 & (\ n \geq 0\ ) \end{cases} \\ &\Rightarrow \quad a_n = 0 \quad \forall n \in \mathbb{Z} \end{aligned}$$

Therefore it also follows that $\dim H_2(\mathcal{E}(B_0,\xi_0,0)) = \dim Ker(\tau(B_0,\xi_0,0)^*) = 0$.

Finally we need to prove that $\dim Ker(\tau(B_0,\xi_0,0)) \cap Ker(\sigma(B_0,\xi_0,0)^*) = \dim W$.
Define $2m \times 2m$ matrices by

$$\begin{aligned} P_n(B_0,\xi_0) &= \begin{pmatrix} (\xi_0^-)_{-n}\ id_{\mathbb{C}^m} & (B_0^+)_{n-1} \\ (\xi_0^+)^*_{-n}\ id_{\mathbb{C}^m} & -(B_0^-)^*_{n-1} \end{pmatrix} \\ Q_n(B_0,\xi_0) &= \begin{pmatrix} (B_0^-)_n & (\xi_0^+)_{-n-1}\ id_{\mathbb{C}^m} \\ (B_0^+)^*_n & -(\xi_0^-)^*_{-n-1}\ id_{\mathbb{C}^m} \end{pmatrix} \end{aligned}$$

Note that they all are invertible since we have

$$\begin{aligned} (n \geq 2) \quad & P_n(B_0,\xi_0) = \begin{pmatrix} in & i(n-1) \\ 0 & \frac{1}{-i(n-1)} \end{pmatrix} \otimes id_{\mathbb{C}^m} & & Q_n(B_0,\xi_0) = \begin{pmatrix} \frac{1}{in} & 0 \\ -in & -i(n+1) \end{pmatrix} \otimes id_{\mathbb{C}^m} \\ (n = 1) \quad & P_1(B_0,\xi_0) = \begin{pmatrix} -i & 1 \\ 0 & 1 \end{pmatrix} \otimes id_{\mathbb{C}^m} & & Q_1(B_0,\xi_0) = \begin{pmatrix} -i & 0 \\ i & -2i \end{pmatrix} \otimes id_{\mathbb{C}^m} \\ (n = 0) \quad & P_0(B_0,\xi_0) = \begin{pmatrix} 0 & -i \\ 1 & -i \end{pmatrix} \otimes id_{\mathbb{C}^m} & & Q_0(B_0,\xi_0) = \begin{pmatrix} 1 & 0 \\ 1 & -i \end{pmatrix} \otimes id_{\mathbb{C}^m} \\ (n \leq -1) \quad & P_n(B_0,\xi_0) = \begin{pmatrix} 0 & \frac{1}{i(n-1)} \\ in & i(n-1) \end{pmatrix} \otimes id_{\mathbb{C}^m} & & Q_n(B_0,\xi_0) = \begin{pmatrix} in & -i(n+1) + \delta_{n,-1} \\ \frac{1}{-in} & 0 \end{pmatrix} \otimes id_{\mathbb{C}^m}. \end{aligned}$$

By the definition of $\sigma$ and $\tau$, we get

$$\left(\sum_{n\in\mathbb{Z}} X_n e_n \otimes e_{-n+1},\ \sum_{n\in\mathbb{Z}} Y_n e_n \otimes e_{-n-1},\ \sum_{n\in I} Z_n e_n \otimes e_{-n}\right) \in Ker\tau(B_0,\xi_0,,0)\cap Ker\sigma(B_0,\xi_0,,0)^*$$
$$\Leftrightarrow \quad P_n(B_0,\xi_0)\begin{pmatrix} X_n \\ Y_{n-1}\end{pmatrix} = Q_n(B_0,\xi_0)\begin{pmatrix} X_{n+1} \\ Y_n\end{pmatrix} \quad \forall n\in\mathbb{Z}.$$

where $I$ denotes the index set of the irreducible decomposition of $W$. From this observation, all of the vectors $\{X_n\}_{n\in\mathbb{Z}},\{Y_n\}_{n\in\mathbb{Z}}$ are uniquely determined up by, for instance, $X_0$ and $Y_{-1}$. If $(X_0,Y_{-1})\neq 0$, then we have $(X_n,Y_n)\neq 0$ for any $n\in\mathbb{Z}$ since the matrices are invertible. Moreover, we prove in the following that $\lim_{n\to\infty}(X_n,Y_n)=\infty$, which indicates that they are not contained in the domain. Therefore we conclude $\dim Ker(\tau(B_0,\xi_0,0))\cap Ker(\sigma(B_0,\xi_0,0)^*)=\dim W$.

In order to prove $\lim_{n\to\infty}(X_n,Y_n)=\infty$, let us consider the matrix for $n>2$,

$$Q_n(B_0,\xi_0)^{-1}P_n(B_0,\xi_0) = \frac{n+1}{n}\begin{pmatrix} -n(n+1) & -(n^2-1) \\ n^2 & n(n-1)-\frac{1}{n(n-1)}\end{pmatrix}\otimes id_{\mathbb{C}^m}$$

Then, we have

$$\begin{aligned}
&{}^t\{Q_n(B_0,\xi_0)^{-1}P_n(B_0,\xi_0)\}Q_n(B_0,\xi_0)^{-1}P_n(B_0,\xi_0)\\
=\ &\left(\frac{n+1}{n}\right)^2\begin{pmatrix} n^2(n+1)^2+n^4 & n(n+1)(n^2-1)+n^3(n-1)-\frac{n}{(n-1)} \\ n(n^2-1)(n+1)+n^3(n-1)-\frac{n}{n-1} & (n^2-1)^2+\left(n(n-1)-\frac{1}{n(n-1)}\right)^2\end{pmatrix}\otimes id_{\mathbb{C}^m}\\
=\ &\left(\frac{n+1}{n}\right)^2\begin{pmatrix} 2n^4+2n^3+n^2 & 2n^4-n^2-n-\frac{n}{n-1} \\ 2n^4-n^2-n-\frac{n}{n-1} & 2n^4-2n^3-2n^2-1+\frac{1}{n^2(n-1)^2}\end{pmatrix}\otimes id_{\mathbb{C}^m}.
\end{aligned}$$

By easy calculations its eigen values $\lambda(n)$ satisfy the equations $2\lambda(n)=A\pm\sqrt{A^2-4B}$, where

$$\begin{aligned}
A &= 4n^4-n^2-1+\frac{1}{n^2(n-1)^2}\ ,\quad A^2=16n^8-8n^6-7n^4+2n^2+O(1)\\
B &= \left(4n^8-6n^6-6n^5-n^4-2n^3-n^2+O(1)\right)\\
&\quad -\left(4n^8-4n^6-4n^5+n^4(1-4\frac{n}{n-1})+2n^3+n^2(1+\frac{2n}{n-1})+2n\frac{n}{n-1}+\frac{n^2}{(n-1)^2}\right)\\
&= -2n^6-2n^5+C_1(n)
\end{aligned}$$

for some polynomial $C_1(n)$ of $n$ whose degree is 4. Thus it follows that

$$A^2-4B=16n^8+8n^5+C_2(n).$$

for some polynomial $C_2(n)$ whose degree is 4. Consequently, we have $\lim_{n\to 0}\lambda(n)=\infty$ and therefore $\lim_{n\to\infty}(X_n,Y_n)=\infty$. □

For a fixed infinite dimensional ADHM data $(V,W,B,\Psi)$, now we have a vector bundle Fredholm complex $\mathcal{E}$ over $X_\zeta$, and we can define a smooth morphism $\mathcal{D}$ by

$$\mathcal{D}\ :\ \mathbb{C}^2\otimes\overline{(C^\infty(S^1,\mathbb{C}^m)\otimes\mathcal{L})^{S^1}}\ \to\ \overline{(C^\infty(S^1,\mathbb{C}^m)\otimes\mathcal{L}\otimes Q)^{S^1}}\oplus(W\otimes\mathcal{L})^{S^1},\ (s,u)\mapsto\sigma(s)+\tau^*(u).$$

What we have seen so far is that $\mathcal{D}$ is a Fredholm operator on each fibre and its index equals $\dim W$, which does not depend on the choice of base points. Now we obtain the following corollary.

**Corollary 4.6.** *For any infinite dimensional ADHM data $(V,W,B,\Psi)$ which satisfies the condition that $\mathcal{D}$ is everywhere injective, $E:=Coker\mathcal{D}\cong Ker\mathcal{D}^*\cong Ker\tau/Im\sigma$ becomes a smooth finite dimensional complex vector bundle and its induced connection is compatible with a complex structure $I$ on $X_\zeta$. In other words, $ker\tau/im\sigma$ becomes a finite dimensional holomorphic vector bundle on $(X_\zeta,I)$.*

*Proof.* The proof directly follows from the previous section with a bit of modification. Let us use the same notation. Since the data is not required to satisfy the real part of the ADHM equation, $\mathcal{D}^*\mathcal{D}$ may not be written as $\mathcal{D}^*\mathcal{D} = id_{\mathbb{C}^2} \otimes \Delta$. However, we still have

$$(\mathcal{D}^*\mathcal{D})_p = \begin{pmatrix} \tilde{\Delta}_p & 0 \\ 0 & \Delta_p \end{pmatrix}$$

due to the complex part of the ADHM equation. It follow from that $\mathcal{D}^*\mathcal{D}$ and $(\mathcal{D}^*\mathcal{D})^{-1}$ are commutable with the complex structure $I = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$ on $\mathbb{C}^2$. Therefore we can derive the equation $R^E(v,w) = R^E(Iv, Iw)$ from exactly the same calculation in the previous section. □

**Remark 4.7.** *There are no infinite dimensional ADHM data which satisfy the real part of the ADHM equation: If we try to satisfy the equation, we must find an element $(A^+, A^-) \in (B^2(L^2(S^1, \mathbb{C}^m)) \otimes Q)^{S^1}$ such that*

$$[\tilde{\Lambda}^+, (A^+)^*] + [A^+, (\tilde{\Lambda}^+)^*] + [\tilde{\Lambda}^-, (A^-)^*] + [A^-, (\tilde{\Lambda}^-)^*] = C(n)\,, \quad on\ \mathbb{C}^m \otimes V_n$$

*for some polynomial $C(n)$ whose degree is exactly 1. But we can not find such an element since it is required that $\lim_{|n|\to\infty} A_n^{\pm} = 0$.*
*On the other hand, we can deliberately define $\tilde{\Lambda}$, for instance $\tilde{\Lambda}^{\pm} = (\Lambda^{\mp})^*$, so that there exist solutions of the real part of the ADHM equation. However, the Fredholm property of the complex $\mathcal{E}$ is no longer guaranteed in these cases. For example, the closedness of the ranges is not guaranteed. Therefore, to the best of our knowledge, we conclude that there exist no instanton constructed from infinite dimensional ADHM data.*

On the other hand, let us fix a complex structure $I$ on $X_\zeta$ and consider $(X_\zeta, I)$ as a complex surface. Since $\sigma$ and $\tau$ are holomorphic maps, we can also obtain the following corollary which is a natural generalization of Corollary 4.6.
**Corollary 4.8.** *For any inifinite dimensional ADHM data, the middle cohomology $E = Ker\tau/Im\sigma$ of the complex $\mathcal{E}$ becomes an analytic sheaf over $X_\zeta$ whose each stalk is finite dimensional.*

# 5 Asymptotic behavior of curvature

In this section we calculate the asymptotic of curvature of instantons constructed from finite dimensional ADHM data. In the following, we regard $\hat{M}, M \subset \mathbb{H}^\infty$ by the inclusion

$$(\xi^+, \xi^-) \longmapsto (Re((\xi^+)_n) + Im((\xi^-)_n)i + Re((\xi^-)_n)j + Im((\xi^-)_n)k)_{n\in\mathbb{Z}}.$$

First, we have a natural $S^1$ action $\tilde{\rho}$ on $\mu^{-1}(\zeta)$ defined by

$$\tilde{\rho} : S^1 \times \mu^{-1}(\zeta) \longrightarrow \mu^{-1}(\zeta)\ ,\ (g,q) \longmapsto \tilde{\rho}(g)(q),$$

$$\tilde{\rho}(g)(q)_n := \begin{cases} q_0 g & (\ n = 0\ ) \\ q_n & (\ n \neq 0\ ) \end{cases}$$

that is right multiplication on $q_0$ with $g \in S^1$. Since $\tilde{\rho}$ is commutative with the action of $G$, it induces the actions on $X_\zeta$. Note that we use the same notation $\tilde{\rho}$ for its induced action on $X_\zeta$. The fixed points of $\tilde{\rho}$ on $X_\zeta$ are given by

$$\left\{\ [(q_n)_{n\in\mathbb{Z}}]\ \middle|\ \exists l \in \mathbb{Z}\ \ s.t.\ q_l = 0\ \right\} \quad \subset \quad \mu^{-1}(\zeta)/G.$$

It can be shown that this subset consists of countable isolated points $\{x_n\}_{n\in\mathbb{Z}}$. (Actually they correspond to the set of intersection points of the rational curves $\{C_n\}_{n\in\mathbb{Z}}$ in $X_\zeta$. For detail, see the next section.) Then, the set of vector fields $\{X^\#, IX^\#, JX^\#, KX^\#\}$ forms a basis of the tangent space at any point on $X^*_\zeta := X_\zeta \backslash \{x_n\}_{n\in\mathbb{Z}}$, where $X^\#$ is the fundamental vector field induced by $X = \sqrt{-1} \in Lie(S^1)$.

Next, let us investigate the tangent vectors at any point of $\pi^{-1}(X^*_\zeta)$, where $\pi : \mu^{-1}(\zeta) \to X_\zeta$ denotes the projection. By the standard argument in the hyper-Kähler quotient [6], we can decompose $T_q\hat{M}$ as

$$T_q\hat{M} = H_q \oplus (T_qG(q) \oplus I(T_qG(q)) \oplus J(T_qG(q)) \oplus K(T_qG(q)))$$

where $T_qG(q)$ is the tangent space of the orbit $G(q)$ at $q \in \hat{M}$, and $H_q = (T_qG(q) \oplus IT_qG(q) \oplus JT_qG(q) \oplus KT_qG(q))^\perp \cong T_{[q]}X_\zeta$. By a straightforward calculation, any vector $(p_n) \in H_q \subset \mathbb{H}^\infty$ satisfies the relation

$$p_n\overline{q_n i} = p_0\overline{q_0 i}.$$

In particular, we have $p_n = p_0 \dfrac{\overline{q_0 i} q_n i}{\|q_n\|^2}$ if $q \in \pi^{-1}(X^*_\zeta)$.

**Lemma 5.1.** *For any tangent vector $Z \in T_q\mu^{-1}(\zeta)$ at $q \in \pi^{-1}(X^*_\zeta)$, we can decompose $Z$ as $Z = Z_H + Z_V$ where $Z_H \in H_q$ and $Z_V \in T_qG(q)$. Then, $Z_H$ is given by $(Z_H)_n = F(q)\tilde{Z}_H \dfrac{\overline{q_0 i} q_n i}{\|q_n\|^2} \in \mathbb{H}$, where $\tilde{Z}_H$ and $F(q)$ is defined by*

$$\tilde{Z}_H \ := \ \sum_{k\in\mathbb{Z}} \sum_{A\in\{1,i,j,k\}} A\left\langle Z_k, A\frac{\overline{q_0 i} q_k i}{\|q_k\|^2}\right\rangle_{\mathbb{R}^4}\ ,\ \ F(q) \ := \ \frac{1}{1 + \sum_{n\neq 0} \frac{\|q_0\|^2}{\|q_n\|^2}}.$$

*Proof.* Because of the equation $(Z_H)_n = (Z_H)_0 \dfrac{\overline{q_0 i} q_n i}{\|q_n\|^2}$, we have only to prove that $(Z_H)_0 = F(q)\tilde{Z}_H$. Since $Z_H$ is given by the projection of $Z$ to $H_q$, $(Z_H)_0$ corresponds to the minimum value of the function $f_Z(p_0) := \|Z - (p_n)\|^2$ of $p_0$. By a simple calculation, we get

$$\begin{aligned} f_Z(p_0) &= \sum_{n\in\mathbb{Z}} \|Z_n - p_n\|^2 \\ &= \sum_{n\in\mathbb{Z}} \|Z_n\|^2 + \|p_n\|^2 - 2 < Z_n,\ p_n >_{\mathbb{R}^4} \\ &= \sum_{n\in\mathbb{Z}} \|Z_n\|^2 + \|p_0\overline{q_0 i}(\overline{q_n i})^{-1}\|^2 - 2 < Z_n,\ p_0\overline{q_0 i}(\overline{q_n i})^{-1} >_{\mathbb{R}^4} \\ &= \sum_{n\in\mathbb{Z}} \|Z_n\|^2 + \frac{\|q_0\|^2}{\|q_n\|^2}\|p_0\|^2 - 2\left\langle Z_n,\ p_0\frac{\overline{q_0 i} q_n i}{\|q_n\|^2}\right\rangle_{\mathbb{R}^4}. \end{aligned}$$

Then its partial differentials are calculated as

$$\frac{\partial}{\partial (p_0)_A} f_Z \quad = \quad 2\sum_{n\in\mathbb{Z}}\left(\frac{\|q_0\|^2}{\|q_n\|^2}(p_0)_A - \left\langle Z_n,\, A\frac{\overline{q_0 i}q_n i}{\|q_n\|^2}\right\rangle_{\mathbb{R}^4}\right), \qquad A\in\{1,i,j,k\},$$

where we write $p_0 = (p_0)_1 + (p_0)_i i + (p_0)_j j + (p_0)_k k \in \mathbb{H}$. Thus, $f_Z$ attains its minimum value at $F(q)\tilde{Z}_H$. □

Now, we have an explicit representation of the natural connection $\theta_\zeta$ on the principal G-bundle $\pi : \mu^{-1}(\zeta) \to X_\zeta$:

$$(\theta_\zeta)_q : T_q\mu^{-1}(\zeta) \longrightarrow T_qG(q) \cong \mathfrak{g}\,, \quad Z \longmapsto Z_V = Z - Z_H.$$

Let us define 1-dimensional representations of G by

$$\rho_k \;:\; G \longrightarrow \mathrm{U}(V_k)\,,\; (g_n)_{n\in\mathbb{Z}} \longmapsto g_k$$

where $G$ as a subgroup of $\times_{n\in\mathbb{Z}}\mathrm{U}(V_n)$, and we get an associated bundle $\mathcal{L}_k := \mu^{-1}(\zeta)\times_{\rho_k}\mathbb{C}$. Since there is the connection $\theta_\zeta$ on $\mu^{-1}(\zeta)$, we have the induced connection $(\theta_\zeta)_k$ on $\mathcal{L}_k$. In order to calculate $(\theta_\zeta)_k$, we need to describe the projection

$$\pi_k : TqG(q) = \mathbb{R}Y_k^\# \oplus (\mathbb{R}Y_k^\#)^\perp \longrightarrow \mathbb{R}Y_k^\#.$$

where $Y_k^\#$ is an fundamental vector field induced by $y_k := \sqrt{-1} \in \mathfrak{u}(V_k) \subset \mathfrak{g}$. As before, $\pi_k(Z)$ corresponds to the minimum value of a function $\tilde{f}(\alpha) := \|Z - \alpha Y_k^\#\|_M^2$, and it follows from an easy calculation that $\tilde{f}(\alpha)$ attains its minimum at

$$\alpha = \frac{g(Y_k^\#, Z)}{\|Y_k^\#\|^2}\,.$$

where $g$ is the Euclidean metric on $\hat{M}$. Therefore, $(\theta_\zeta)_k$ is expressed as a $\mathfrak{u}(V_k)$-valued 1-form on $\mu^{-1}(\zeta)$:

$$(\theta_k)_q = d(\rho_k)_e \circ \pi_k \circ \theta_q \;:\; T_q\mu^{-1}(\zeta) \to T_qG(q) \cong \mathfrak{g} \to \mathfrak{u}(V_k), \quad Z \longmapsto \frac{g(Y_k^\#, Z_V)}{\|Y_k^\#\|^2}\sqrt{-1}$$

It is clear from the definition that $\|Y_k^\#\|^2 = \|q_k\|^2 + \|q_{k-1}\|^2$.

Now we have the following estimation of the norm of the curvature $(R_\zeta)_k$ of $(\theta_\zeta)_k$ on $X_\zeta^*$:

$$\begin{aligned}
\|(R_\zeta)_k\| \quad \leq \quad & \frac{1}{\|Y_k^\#\|^2}\left(\frac{|g(Y_k^\#,[\tilde{X}^\#, I\tilde{X}^\#])|}{\|\tilde{X}^\#\|\|I\tilde{X}^\#\|} + \frac{|g(Y_k^\#,[\tilde{X}^\#, J\tilde{X}^\#])|}{\|\tilde{X}^\#\|\|J\tilde{X}^\#\|} + \frac{|g(Y_k^\#,[\tilde{X}^\#, K\tilde{X}^\#])|}{\|\tilde{X}^\#\|\|K\tilde{X}^\#\|}\right.\\
& \left. + \frac{|g(Y_k^\#,[I\tilde{X}^\#, J\tilde{X}^\#])|}{\|I\tilde{X}^\#\|\|J\tilde{X}^\#\|} + \frac{|g(Y_k^\#,[I\tilde{X}^\#, K\tilde{X}^\#])|}{\|I\tilde{X}^\#\|\|K\tilde{X}^\#\|} + \frac{|g(Y_k^\#,[J\tilde{X}^\#, K\tilde{X}^\#])|}{\|J\tilde{X}^\#\|\|K\tilde{X}^\#\|}\right)\\
= \quad & \frac{1}{\|Y_k^\#\|^2\|\tilde{X}^\#\|^2}\Big(|g(Y_k^\#,[\tilde{X}^\#, I\tilde{X}^\#])| + |g(Y_k^\#,[\tilde{X}^\#, J\tilde{X}^\#])| + |g(Y_k^\#,[\tilde{X}^\#, K\tilde{X}^\#])|\\
& + |g(Y_k^\#,[I\tilde{X}^\#, J\tilde{X}^\#])| + |g(Y_k^\#,[I\tilde{X}^\#, K\tilde{X}^\#])| + |g(Y_k^\#,[J\tilde{X}^\#, K\tilde{X}^\#])|\Big).
\end{aligned}$$

where $\tilde{X}^\#$ denotes the horizontal lift of $X^\#$ to $\mu^{-1}(\zeta)$, and we use the fact that $(R_\zeta)_k(v,w) = -(\theta_\zeta)_k([\tilde{v},\tilde{w}])$.

**Remark 5.2.** *In terms of the quaternion, n-th component of the moment map can be written as*

$$\mu(q)_n = \frac{1}{2}\Big(q_{n-1}i\,\overline{q_{n-1}} - q_n i\,\overline{q_n} - \Lambda_{n-1}i\overline{\Lambda}_{n-1} + \Lambda_n i\overline{\Lambda}_n\Big).$$

*Thus, we have*

$$\begin{aligned}
\|q_n\|^2 \quad &\leq \quad \|q_0\|^2 + n^2 + 2\left\|\sum_{k=0}^{k=n}\zeta_k\right\| \leq \|q_0\|^2 + n^2 + 2|n|\|\zeta\|_{\mathfrak{g}\otimes Im\mathbb{H}}\\
\|q_0\|^2 \quad &\leq \quad \|q_n\|^2 + n^2 + 2|n|\|\zeta\|_{\mathfrak{g}\otimes Im\mathbb{H}}
\end{aligned}$$

*on $\mu^{-1}(\zeta)$. This shows that $O(\|q_n\|) = O(\|q_0\|)(\ \|q_0\| \to \infty\ )$ and therefore we have only to check the order of a function with respect to $\|q_0\|$ in order to investigate its asymptotical behavior on $\mu^{-1}(\zeta)$.*

**Lemma 5.3.** *For any fixed $k \in \mathbb{Z}$, we have the equation $\|(R_\zeta)_k\| = O(\|q_0\|^{-5})$ on $X_\zeta$. In other words, there exist two positive constants $C_1, C_2 \in \mathbb{R}$, which depend only on $k$ and $\zeta$, such that $\|(R_\zeta)_k\| \le C_1\|q_0\|^{-5}$ when $C_2 \le \|q_0\|$.*

*Proof.* For simplicity, we only prove the case in which $\zeta = \mu(\Lambda) = 0$. Other cases follow with a slight modification. Since $\{x_n\}_{n\in\mathbb{Z}}$ are isolated, it is sufficient that we regard $\|(R_\zeta)_k\|$ as a function on $\pi^{-1}(X^*_\zeta)$ and estimate its order.
To begin with, we need to determine $(\tilde{X^\#})_q$ for all $q \in \pi^{-1}(X^*_\zeta)$. We denote by $W$ the vector field induced by $X = \sqrt{-1} \in Lie(S^1)$ through the action $\tilde{\rho}$ on $\mu^{-1}(\zeta)$. More explicitly, $(W)_q \in T_q\mu^{-1}(\zeta) \subset \mathbb{H}^\infty$ is given by

$$((W)_q)_n = \begin{cases} q_0 i & (n = 0) \\ 0 & (n \neq 0) \end{cases}.$$

Since $(\tilde{X^\#})_q = (W_q)_H$, we have $((\tilde{X^\#})_q)_n = F(q)\dfrac{\|q_0\|^2}{\|q_n\|^2} q_n i$ by Lemma 5.1. In addition, we see that

$$((I\tilde{X^\#})_q)_n = F(q)\frac{\|q_0\|^2}{\|q_n\|^2} i q_n i, \qquad ((J\tilde{X^\#})_q)_n = F(q)\frac{\|q_0\|^2}{\|q_n\|^2} j q_n i, \qquad ((K\tilde{X^\#})_q)_n = F(q)\frac{\|q_0\|^2}{\|q_n\|^2} k q_n i.$$

For any $A \in \{1, I, J, K\}$, we define $q^A_t$ as the integral curve for $A\tilde{X^\#}$ which is given by $q^A_t = q + tA\tilde{X^\#} + O(t^2) \in \mu^{-1}(q)$ near $q \in \mu^{-1}(\zeta)$. Then, let us take any $B \in \{1, I, J, K\}$ and we have

$$\begin{aligned} ([A\tilde{X^\#}, B\tilde{X^\#}]_q)_n &= \left.\frac{d}{dt}\right|_{t=0} F(q^A_t)\frac{\|(q^A_t)_0\|^2}{\|(q^A_t)_n\|^2} B(q^A_t)_n i \\ &= \left(\left.\frac{d}{dt}\right|_{t=0} F(q^A_t)\frac{\|(q^A_t)_0\|^2}{\|(q^A_t)_n\|^2}\right) B q_n i + F(q)\frac{\|q_0\|^2}{\|q_n\|^2}\left(\left.\frac{d}{dt}\right|_{t=0} B(q^A_t)_n i\right). \end{aligned}$$

Furthermore, we get

$$\begin{aligned} g_q(Y^\#_k,\ [A\tilde{X^\#}, B\tilde{X^\#}]) &= < q_k i,\ ([A\tilde{X^\#}, B\tilde{X^\#}]_q)_k >_{\mathbb{R}^4} + < -q_{k-1} i,\ ([A\tilde{X^\#}, B\tilde{X^\#}]_q)_{k-1} >_{\mathbb{R}^4} \\ &= Re\Big(([A\tilde{X^\#}, B\tilde{X^\#}]_q)_k\ \overline{q_k i}\Big) - Re\Big(([A\tilde{X^\#}, B\tilde{X^\#}]_q)_{k-1}\ \overline{q_{k-1} i}\Big) \\ &= F(q)\frac{\|q_0\|^2}{\|q_k\|^2} Re\left(\left.\frac{d}{dt}\right|_{t=0} B(q^A_t)_k i\ \overline{q_k i}\right) - F(q)\frac{\|q_0\|^2}{\|q_{k-1}\|^2} Re\left(\left.\frac{d}{dt}\right|_{t=0} B(q^A_t)_{k-1} i\ \overline{q_{k-1} i}\right) \\ &= F(q)\frac{\|q_0\|^2}{\|q_k\|^2} Re\left(B\left(F(q)\frac{\|q_0\|^2}{\|q_k\|^2} A q_k i\right) i\ \overline{q_k i}\right) - F(q)\frac{\|q_0\|^2}{\|q_{k-1}\|^2} Re\left(B\left(F(q)\frac{\|q_0\|^2}{\|q_{k-1}\|^2} A q_{k-1} i\right) i\ \overline{q_{k-1} i}\right) \\ &= -F(q)^2\frac{\|q_0\|^4}{\|q_k\|^4} Re\Big(BAq_k\overline{q_k i}\Big) + F(q)^2\frac{\|q_0\|^4}{\|q_{k-1}\|^4} Re\Big(BAq_{k-1}\overline{q_{k-1} i}\Big) \\ &= F(q)^2\|q_0\|^4 Re\left[BA\left(\frac{q_k i\ \overline{q_k}}{\|q_k\|^4} - \frac{q_{k-1} i\ \overline{q_{k-1}}}{\|q_{k-1}\|^4}\right)\right] \\ &= F(q)^2\|q_0\|^4 Re\left[BA\left(\frac{q_{k-1} i\ \overline{q_{k-1}} + C_1}{\|q_k\|^4} - \frac{q_{k-1} i\ \overline{q_{k-1}}}{\|q_{k-1}\|^4}\right)\right] \\ &= F(q)^2\frac{\|q_0\|^4}{\|q_k\|^4} C_2 + F(q)^2\|q_0\|^4 Re\left[BA\left(\frac{1}{\|q_k\|^4} - \frac{1}{\|q_{k-1}\|^4}\right) q_{k-1} i\ \overline{q_{k-1}}\right] \end{aligned}$$

where $C_1, C_2$ are constants which depend on $k$ and $\zeta$, and we use $\left.\dfrac{d}{dt}\right|_{t=0} (q^A_t)_n = (A\tilde{X^\#})_n = F(q)\dfrac{\|q_0\|^2}{\|q_n\|^2} A q_n i$.
In addition, it follows from Remark 5.2 that $q_k i\ \overline{q_k} = q_{k-1} i\ \overline{q_{k-1}} + C_3$ for a constant $C_3 \in Im\mathbb{H}$ which depends

on $k$ and $\zeta$. By taking the square of the both sides, we have

$$\begin{aligned}
& (q_k\overline{iq_k})^2 = (q_{k-1}\overline{iq_{k-1}})^2 + q_{k-1}\overline{iq_{k-1}}C_3 + Cq_{k-1}\overline{iq_{k-1}} + C_3^2 \\
\Leftrightarrow \quad & -\|q_k\|^4 = -\|q_{k-1}\|^4 + q_{k-1}\overline{iq_{k-1}}C_3 + C_3q_{k-1}\overline{iq_{k-1}} + C_3^2 \\
\Leftrightarrow \quad & \frac{1}{\|q_k\|^4} - \frac{1}{\|q_{k-1}\|^4} = \frac{q_{k-1}\overline{iq_{k-1}}C_3 + C_3q_{k-1}\overline{iq_{k-1}} + C_3^2}{\|q_k\|^4\|q_{k-1}\|^4} = O(\|q_0\|^{-6}).
\end{aligned}$$

As a result, we get

$$\left|g(Y_k^{\#}, [A\tilde{X}^{\#}, B\tilde{X}^{\#}])\right| = F(q)^2O(1).$$

Finally, we need to calculate $\|\tilde{X}^{\#}\|^2$ and estimate the order of $F(q)$. It follows from the definition that

$$\begin{aligned}
\|(\tilde{X}^{\#})_n\|^2 &= \left\|F(q)\frac{\|q_0\|^2}{\|q_n\|^2}q_n i\right\|^2 = F(q)^2\frac{\|q_0\|^4}{\|q_n\|^2}, \\
\|\tilde{X}^{\#}\|^2 &= \sum_{n\in\mathbb{Z}}\|(\tilde{X}^{\#})_n\|^2 = F(q)^2\|q_0\|^2\left(1+\sum_{n\neq 0}\frac{\|q_0\|^2}{\|q_n\|^2}\right) = F(q)\|q_0\|^2.
\end{aligned}$$

In order to calculate the order of $F(q)$, we see that

$$1+\sum_{n\neq 0}\frac{\|q_0\|^2}{\|q_n\|^2} \geq 1+\sum_{n\neq 0}\frac{\|q_0\|^2}{\|q_0\|^2+n^2} = 1+\|q_0\|^2\left(\frac{\pi\coth(\pi\|q_0\|)}{\|q_0\|} - \frac{1}{\|q_0\|^2}\right) = \pi\|q_0\|\coth(\pi\|q_0\|) \geq 0$$

for sufficiently large $\|q_0\|$. Therefore, we have $F(q) = O(\|q_0\|^{-1})$. Note that this is the only point which we have to slightly modify if we prove the case for general $\zeta$.

Summarizing the above, we can calculate the order of $\frac{1}{\|Y_k^{\#}\|^2\|\tilde{X}^{\#}\|^2}\,|g(Y_k^{\#}, [A\tilde{X}^{\#}, B\tilde{X}^{\#}])|$:

$$\begin{aligned}
\frac{1}{\|Y_k^{\#}\|^2\|\tilde{X}^{\#}\|^2}\,|g(V_k^{\#}, [A\tilde{X}^{\#}, B\tilde{X}^{\#}])| &= \frac{1}{\|q_k\|^2+\|q_{k-1}\|^2}\frac{1}{F(q)\|q_0\|^2}F(q)^2O(1) \\
&= F(q)O(\|q_0\|^{-4}) \\
&= O(\|q_0\|^{-5}),
\end{aligned}$$

and we conclude that $\|(R_\zeta)_k\| = O(\|q_0\|^{-5})$. □

**Remark 5.4.** *In the case of hyper-Kähler manifolds of type $A_k$ in which $\hat{M} \cong \mathbb{H}^{k+1}$ and F(q)=O(1), it follows from the above calculation that $\|(R_\zeta)_k\| = O(\|q_0\|^{-4})$. This is the result given in [11]. We can also induce from the following proof of the Proposition 5.5 that $\|R^E\| = O(\|q_0\|^{-4})$ for any finite dimensional ADHM data in the case of hyper-Kähler manifolds of type $A_k$.*

**Proposition 5.5.** *For any finite ADHM data which satisfies the condition that $\mathcal{D}$ is everywhere injective, the norm of the curvature $R^E$ of the induced connection on $E = Coker\mathcal{D}$ equals $O(\|q_0\|^{-5})$. In other words, there exist two positive constants $C_1, C_2 \in \mathbb{R}$, which depend only on the ADHM data and $\zeta$, such that $\|R^E\| \leq C_1\|q_0\|^{-5}$ when $C_2 \leq \|q_0\|$.*

*Proof.* According to the proof of Theorem 3.5, $R^E(v,w) \in \Gamma(End(E))$ is written as

$$\begin{aligned}
R^E(v,w) = \pi_E \circ \Big(& R(v,w) + \pi_Q^* \circ \nabla_v(id_V \otimes \xi) \circ H \circ (\nabla_w(id_V \otimes \xi))^* \circ \pi_Q \\
& - \pi_Q^* \circ \nabla_w(id_V \otimes \xi) \circ H \circ (\nabla_v(id_V \otimes \xi))^* \circ \pi_Q\Big)\Big|_E
\end{aligned}$$

where $v$ and $w$ are any tangent vectors on $X_\zeta$. In this case, the restriction of $\|R\|$ on $E$ is bounded by a finite sum of $\|(R_\zeta)_k\|$s. Therefore by Lemma 5.2, we have only to consider the terms coming from the second fundamental form. As before, it is enough to calculate on $X_\zeta^*$.

First, let us decompose $V$ and $W$ as $V = \oplus_{i\in S_V}\mathbb{C}^{n_i}\otimes V_i$ and $W = \oplus_{i\in S_W}\mathbb{C}^{m_i}\otimes V_i$, where $S_V$ and $S_W$ are finite index sets. For a fixed base point $[q]\in X^*_\zeta$, we can write the fibre of the vector bundles as

$$\begin{aligned}\left((V\otimes\mathcal{L}\otimes Q)^{S^1}\oplus(W\otimes\mathcal{L})^{S^1}\right)_{[q]} &= \left\{\sum_{k\in S_V} a_k e_k\otimes e_{-k+1} \mid a_k\in\mathbb{C}^{n_k}\right\}\oplus\left\{\sum_{k\in S_V} b_k e_k\otimes e_{-k-1} \mid b_k\in\mathbb{C}^{n_k}\right\}\\ &\quad\oplus\left\{\sum_{k\in S_W} c_k e_k\otimes e_{-k} \mid c_k\in\mathbb{C}^{m_k}\right\},\\ \left((V\otimes\mathcal{L})^{S^1}\right)_{[q]} &= \left\{\sum_{k\in S_V} a_k e_k\otimes e_{-k} \mid a_k\in\mathbb{C}^{n_k}\right\}.\end{aligned}$$

Using the matrix expression as in section 3, we have

$$\nabla_w(id_V\otimes\xi)_{[q]}\begin{pmatrix} e_n\otimes e_{-n}\\ e_m\otimes e_{-m}\end{pmatrix} = e_n\otimes\begin{pmatrix} d\xi^+(\tilde{w})_q(e_{-n})\\ d\xi^-(\tilde{w})_q(e_{-n})\end{pmatrix}+e_m\otimes\begin{pmatrix}(d(\xi^-)_q(\tilde{w}))^*(e_{-m})\\ -(d(\xi^+)_q(\tilde{w}))^*(e_{-m})\end{pmatrix}$$

where $(d\xi^+(\tilde{w})_q, d\xi^-(\tilde{w})_q)\in T_q\hat{M} = (B^2(S^1)\otimes Q)^{S^1}\subset\mathbb{H}^\infty$ is the differential of the canonical coordinate function $\xi$ of $\hat{M}$ along $\tilde{w}$. Therefore if we take $w$ to be $AX^\#$ where $A\in\{1,I,J,K\}$, it follows that

$$\nabla_{AX^\#}(id_V\otimes\xi)_{[q]} = F(q)O(\|q_0\|)$$

since we have $\|(\tilde{X^\#})_n\|^2 = F(q)^2\frac{\|q_0\|^4}{\|q_n\|^2}$ for any $n\in\mathbb{Z}$.
Next, we know $H$ is defined by $H = (\mathcal{D}^*\mathcal{D})^{-1} = (id_{\mathbb{C}^2}\otimes\Delta)^{-1} = id_{\mathbb{C}^2}\otimes\Delta^{-1}$ where $\Delta$ is written as

$$\begin{aligned}\Delta &= \frac{1}{2}\left(\sum_{l\in\{+,-\}}\left(B^i(B^i)^*+(B^i)^*B^i\right)+\Psi_1(\Psi_1)^*+(\Psi_2)^*\Psi_2\right)\otimes id_{\mathcal{L}}\\ &\quad + id_V\otimes\frac{1}{2}\left(\xi^+(\xi^+)^*+\xi^-(\xi^-)^*+(\xi^+)^*\xi^++(\xi^-)^*\xi^-\right)\\ &\quad+\sum_{l\in\{+,-\}}\left((B^i)^*\otimes\xi^i+B^i\otimes(\xi^i)^*\right),\end{aligned}$$

and it follows that $H = O(\|q_0\|^{-2})$. Therefore we have

$$\begin{aligned}\nabla_{AX^\#}(id_V\otimes\xi)\circ H\circ(\nabla_{BX^\#}(id_V\otimes\xi))^* &= F(q)O(\|q_0\|)O(\|q_0\|^{-2})F(q)O(\|q_0\|)\\ &= F(q)^2O(1)\end{aligned}$$

for any $A,B\in\{1,I,J,K\}$. Moreover, as in the proof of Theorem 4.1 in [11], it follows that

$$\pi_Q\big|_E = O(\|q_0\|^{-1}).$$

Summarizing above, we have

$$\begin{aligned}\left\|\pi_E\circ\pi_Q^*\circ\nabla_{X^\#}(id_V\otimes\xi)\circ H\circ(\nabla_{IX^\#}(id_V\otimes\xi))^*\circ\pi_Q\big|_E\right\| &= O(\|q_0\|^{-1})F(q)^2O(1)O(\|q_0\|^{-1})\\ &= F(q)^2O(\|q_0\|^{-2})\end{aligned}$$

and therefore we conclude

$$\begin{aligned}\|R^E\| &\le \|R\|+6\|X^\#\|^{-2}\left\|\pi_E\circ\pi_Q^*\circ\nabla_{X^\#}(id_V\otimes\xi)\circ H\circ(\nabla_{IX^\#}(id_V\otimes\xi))^*\circ\pi_Q\big|_E\right\|\\ &= O(\|q_0\|^{-5})+\frac{1}{F(q)\|q_0\|^2}F(q)^2O(\|q_0\|^{-2})\\ &= O(\|q_0\|^{-5}).\end{aligned}$$

□

**Remark 5.6.** *According to the proof of proposition 5.5, the constants $C_1, C_2$ in the statement depend on the number of elements in $S_V$. Therefore we can "not" directly generalize the result to the cases with infinite dimensional data.*

# 6 Topological property of the line bundle $\mathcal{L}_n$

In this section, we review the fact that $H_2(X_\zeta,\mathbb{Z})$ is in the form of the Dynkin diagram of type $A_\infty$ [6]:

$$\cdots \;-\; \underset{C_{-k}}{\circ} \;-\; \underset{C_{-k+1}}{\circ} \;-\; \cdots \;-\; \underset{C_{-2}}{\circ} \;-\; \underset{C_{-1}}{\circ} \;-\; \underset{C_{0}}{\circ} \;-\; \underset{C_{1}}{\circ} \;-\; \underset{C_{2}}{\circ} \;-\; \cdots \;-\; \underset{C_{k-1}}{\circ} \;-\; \underset{C_{k}}{\circ} \;-\; \cdots$$

where each circle denotes a rational curve $\mathbb{C}P^1$ whose self intersection number is $-2$, and two circles are joined by a line segment whenever the corresponding curves intersect. In other words, the middle homology of $X_\zeta$ is generated by the rational curves $\{C_n\}_{n\in\mathbb{Z}}$ which are homeomorphic to $\mathbb{C}P^1$ and its intersection form is given by the Cartan matrix of type $A_\infty$:

$$< C_n, C_m >= \begin{cases} -2 & (\, n = m \,) \\ 1 & (\, |n-m| = 1 \,) \\ 0 & (otherwise) \end{cases}$$

After we see the above fact, we prove that the first Chern class of $\mathcal{L}_n$ is the dual of $C_n$, and therefore $\{c_1(\mathcal{L}_n)\}_{n\in\mathbb{Z}}$ forms a basis of the middle cohomology $H^2(X_\zeta,\mathbb{Z})$.

First, we again consider the $S^1$-action $\tilde{\rho}$ on $X_\zeta$ as in the previous section, that is given by

$$\tilde{\rho} \;:\; S^1 \times X_\zeta \longrightarrow X_\zeta, \;\; (\, g, \, [(q_n)] \,) \longmapsto [\; q_0 g \;, \; (q_n)_{n\in\mathbb{Z}\setminus\{0\}} \;].$$

Then its hyper-Kähler moment map, which depends on $\zeta$, is given by

$$\mu_\zeta(\; [(q_n)] \;) := \frac{1}{2} q_0 i \overline{q}_0 \quad \in Im\mathbb{H}.$$

By Remark 5.2, we have the value of $\mu_\zeta$ in terms of $q_n$ as

$$\mu_\zeta(\; [(q_n)] \;) = \frac{1}{2} q_0 i \overline{q_0} = \frac{1}{2} \begin{cases} q_n i \overline{q}_n - \Lambda_n i \overline{\Lambda}_n + 2 \displaystyle\sum_{k=0}^{n} \zeta_k & (n \geq 1) \\ q_n i \overline{q}_n - \Lambda_n i \overline{\Lambda}_n - 2 \displaystyle\sum_{k=n+1}^{0} \zeta_k & (n \leq -1) \end{cases} .$$

As we mention in the previous section, it is easy to see that $S^1$ acts freely on

$$X^*_\zeta \;=\; \left\{ \; [(q_n)_{n\in\mathbb{Z}}] \;\middle|\; q_n \neq 0 \,,\, n \in \mathbb{Z} \; \right\} \subset \mu^{-1}(\zeta)/G \;=\; X_\zeta$$

and its fixed points are isolated. Moreover, the image of the fixed points $\{\lambda(\zeta)_n\}_{n\in\mathbb{Z}}$ under $\mu_\zeta$ is given by

$$\lambda(\zeta)_n = \mu_\zeta(\; [\cdots, q_{n-1}, 0, q_{n+1}, \cdots] \;) = \frac{1}{2} \begin{cases} -\Lambda_n i \overline{\Lambda}_n + 2 \displaystyle\sum_{k=0}^{n} \zeta_k & (n \geq 1) \\ 0 & (n = 0) \\ -\Lambda_n i \overline{\Lambda}_n - 2 \displaystyle\sum_{k=n+1}^{0} \zeta_k & (n \leq -1) \end{cases} .$$

Note that, $\lambda(\zeta)_n \neq \lambda(\zeta)_m$ for $n \neq m$ since $\zeta$ is generic. If we put $Y_\zeta := Im\mathbb{H}\setminus\{\lambda(\zeta)_n\}$, we have the principal $S^1$-bundle $\mu_\zeta : X^*_\zeta \to Y_\zeta$.

For any $q_1, q_2 \in Im\mathbb{H}$, we define

$$\begin{aligned} [q_1, q_2] \;&:=\; \{\, (1-t)q_1 + t q_2 \mid t \in [0,1] \,\} \subset Im\mathbb{H} \\ (q_1, q_2) \;&:=\; \{\, (1-t)q_1 + t q_2 \mid t \in (0,1) \,\} \subset Im\mathbb{H} \end{aligned}$$

and put $C_{n,m} := \mu_\zeta^{-1}([\lambda(\zeta)_n, \lambda(\zeta)_m])$. If we take $n, m$ such that $(\lambda(\zeta)_n, \lambda(\zeta)_m) \subset Y_\zeta$, then we have $C_{n,m} \cong \mathbb{C}P^1$ as a smooth manifold, whose orientation is defined as follows:
Suppose that the principal $S^1$ bunddle $\mu_\zeta^{-1}((\lambda(\zeta)_n, \lambda(\zeta)_m)) \to (\lambda(\zeta)_n, \lambda(\zeta)_m)$ is trivial and let $\sigma$ be a trivialization. Then, we take a coordinate $(t, \theta)$ given by

$$\begin{aligned} t &: \mu_\zeta^{-1}((\lambda(\zeta)_n, \lambda(\zeta)_m)) \longrightarrow \mathbb{R}\ ,\ \sigma((1-t)\lambda(\zeta)_n + t\lambda(\zeta)_m)e^{iu} \longmapsto t, \\ u &: \mu_\zeta^{-1}((\lambda(\zeta)_n, \lambda(\zeta)_m)) \longrightarrow \mathbb{R}\ ,\ \sigma((1-t)\lambda(\zeta)_m + t\lambda(\zeta)_m)e^{iu} \longmapsto u, \end{aligned}$$

and define the orientation of $C_{n,m}$ by $dt \wedge du$.
By definition, $C_{n,m}$ and $C_{m,n}$ have the opposite orientations and therefore it is easy to see that

$$[C_{n,m}] + [C_{m,l}] + [C_{l,n}] = 0$$

for any $n, m, l \in \mathbb{Z}$, where we denote by $[C_{n,m}]$ the homology class of $C_{n,m}$.

**Lemma 6.1** ([6],[8]). *Let us take an appropriate bijection $\psi : \mathbb{Z} \to \mathbb{Z}$ so that $(\lambda(\zeta)_{n-1}, \lambda(\zeta)_n) \subset Y_\zeta$ for all $n \in \mathbb{Z}$. Then there exists a deformation retract of $\bigcup_{n\in\mathbb{Z}} C_{n-1,n}$.*

*Proof.* There exists a deformation retract $F : Im\mathbb{H} \times [0,1] \to Im\mathbb{H}$ which satisfies $F(-,0) = id_{Im\mathbb{H}}$, $F(Im\mathbb{H}, 1) = \bigcup_{n\in\mathbb{Z}} .[\lambda(\zeta)_n, \lambda(\zeta)_{n+1}]$ and $F(\eta, 1) = \eta$ $(\forall \eta \in \bigcup_{n\in\mathbb{Z}}[\lambda(\zeta)_n, \lambda(\zeta)_{n+1}])$. Now we can take the horizontal lift $\tilde{F} : X_\zeta \times [0,1] \to X_\zeta$ of $F$ by using a $S^1$-connection, and this is the required deformation retract. □

**Corollary 6.2** ([6],[8]). *By taking an appropriate bijection $\psi : \mathbb{Z} \to \mathbb{Z}$ so that $(\lambda(\zeta)_{n-1}, \lambda(\zeta)_n) \subset Y_\zeta$ for all $n \in \mathbb{Z}$, $H_2(X_\zeta, \mathbb{Z})$ is generated by the rational curves $\{C_{n-1,n}\}$ which are homeomorphic to $\mathbb{C}P^1$, and its intersection form is expressed by the Cartan matrix of type $A_\infty$.*

On the other hand, we now have a free tri-Hamiltonian action on the hyper-Kähler manifold $X_\zeta^*$ in the sense of the following definition:
**Definition 6.3** ([8]). *An $S^1$-action on a 4-dimensional hyper-Kähler manifold $(X, \omega)$ is tri-Hamiltonian if the action preserves $\omega$ and there exists a hyper-Kähler moment map $\mu \to Im\mathbb{H}$ with respect to the $S^1$-action.*

For the class of free tri-Hamiltonian actions on hyper-Kähler 4-manifolds, we have the following useful and well-known fact:
**Theorem 6.4** ([6],[8]). *There exists the one-to-one correspondence between the followings:*
*(i) a hyper-Kähler 4-manifold with a free tri-Hamiltonian $S^1$ action.*
*(ii) a principal $S^1$-bundle $\mu : X \to Y$ with an $S^1$-connection $A$ on $X$, where $Y$ is an open subset of $\mathbb{R}^3$, and a positive harmonic function $\Phi$ on $Y$ such that $dA = \mu^*(*d\Phi)$.*
*Here we denote by $*$ the Hodge star operator with respect to the Euclidean metric on $\mathbb{R}^3$.*

According to the proof of the Theorem 6.4, the positive harmonic function $\Phi_\zeta$ corresponding to the principal $S^1$-bundle $\mu_\zeta : X_\zeta^* \to Y_\zeta \subset Im\mathbb{H} \cong \mathbb{R}^3$ is given by,

$$\Phi_\zeta(\eta) = \sum_{n\in\mathbb{Z}} \frac{1}{\|\eta - \lambda(\zeta)_n\|}$$

for $\eta \in Y_\zeta$ and its hyper-Kähler structure $\omega_I, \omega_J, \omega_K$ on $X_\zeta^*$ can be written as

$$\begin{aligned} \omega_I &= A \wedge d(\mu_\zeta)_1 + (\mu_\zeta)^*\Phi_\zeta d(\mu_\zeta)_2 \wedge d(\mu_\zeta)_3\ , \\ \omega_J &= A \wedge d(\mu_\zeta)_2 + (\mu_\zeta)^*\Phi_\zeta d(\mu_\zeta)_3 \wedge d(\mu_\zeta)_1\ , \\ \omega_K &= A \wedge d(\mu_\zeta)_3 + (\mu_\zeta)^*\Phi_\zeta d(\mu_\zeta)_1 \wedge d(\mu_\zeta)_2\ , \end{aligned}$$

where we put $\mu_\zeta = (\mu_\zeta)_1 i + (\mu_\zeta)_2 j + (\mu_\zeta)_3 k \in Im\mathbb{H}$.

Now we are ready to prove that the first Chern classes $\{c_1(\mathcal{L}_n)\}$ form a basis of $H^2(X_\zeta, \mathbb{Z})$.
**Proposition 6.5.** *Let us take an appropriate bijection $\psi : \mathbb{Z} \to \mathbb{Z}$ so that $(\lambda(\zeta)_{n-1}, \lambda(\zeta)_n) \subset Y_\zeta$ for all $n \in \mathbb{Z}$. Then the first Chern classes $\{c_1(\mathcal{L}_n)\}$ form a dual basis of $\{C_{n-1,n}\}_{n\in\mathbb{Z}}$. In other words, we have*

$$\langle c_1(\mathcal{L}_n), [C_{m-1,m}] \rangle = -\frac{1}{2\pi} \int_{C_{m-1,m}} (R_\zeta)_n = \delta_{n,m}.$$

*Proof.* We may assume that $\mu_\zeta^{-1}((\lambda(\zeta)_{m-1},\lambda(\zeta)_m)) \to (\lambda(\zeta)_{m-1},\lambda(\zeta)_m)$ is trivial without a loss of generality. Let us take a coordinate $(t,u)$ on $\mu_\zeta^{-1}((\lambda(\zeta)_{m-1},\lambda(\zeta)_m))$ as above, and define

$$\gamma_\zeta(t) \quad := \quad (1-t)\lambda(\zeta)_{m-1} \ + \ t\lambda(\zeta)_m \in Im\mathbb{H}$$

for $t \in [0,1]$. Then the homology class $[C_{m-1,m}]$ is represented by $\mu^{-1}(\gamma_\zeta([0,1]))$. Moreover, the pullback of the hyper-Kähler form $\omega_\zeta = (\omega_\zeta)_1 i + (\omega_\zeta)_2 j + (\omega_\zeta)_3 k$ on the coordinate $(t,u)$ is written as

$$\omega_\zeta = -(\lambda(\zeta)_m - \lambda(\zeta)_{m-1})\,dt \wedge du, \qquad on \ \ \mu_\zeta^{-1}((\lambda(\zeta)_{m-1},\lambda(\zeta)_m)).$$

In addition, by Duistermaat-Heckman theorem [4], we have

$$< \ v, \ R_\zeta \ >= \frac{\partial}{\partial v}\omega_\zeta \qquad on \ \ \mu_\zeta^{-1}((\lambda(\zeta)_{m-1},\lambda(\zeta)_m)).$$

where $v$ is a finite linear combination of $\{y_k \otimes i, y_k \otimes J, y_k \otimes K\}_{k\in\mathbb{Z}} \in \mathfrak{g} \otimes Im\mathbb{H}$, where $y_k = \sqrt{-1} \in \mathfrak{u}(V_k) \subset \mathfrak{g}$. Note that we denote by $< \, , \, >$ the formal metric on $\oplus_{k\in\mathbb{Z}}\mathfrak{u}(V_k)$, which is calculated as a sum of the inner product on each $V_k$. Since we are working on the infinite dimensional hyper-Kähler quotient setting, the finiteness of the linear combination is necessary. (The proof of Duistermaat-Heckman theorem [4] only works for finite vectors in $\mathfrak{g} \otimes Im\mathbb{H}$).

Then, taking $v$ to be $v_n \otimes i + v_n \otimes j + v_n \otimes k$, it follows that

$$\begin{aligned}(R_\zeta)_n \otimes (i+j+k) \ = \ < \ v, \ R_\zeta \ > \ &= \ -\left(\frac{\partial}{\partial v}\zeta_m\right) dt \wedge du \\ &= \ -\delta_{n,m} dt \wedge du \otimes (i+j+k).\end{aligned}$$

Therefore we conclude

$$\begin{aligned}-\frac{1}{2\pi}\int_{C_{m-1,m}} (R_\zeta)_n \ &= \ \frac{\delta_{n,m}}{2\pi}\int_0^{2\pi}\int_0^1 dt \wedge du \\ &= \ \delta_{n,m}.\end{aligned}$$

□

# 7 $L^p$-norms of instantons

In this section, we prove that the norms of the curvature $R^E$ of instantons constructed from finite dimensional ADHM data are $L^p$-integrable for $p > \frac{6}{5}$. First we prove the finiteness when $p = 2$, and then generalize it.

According to the previous section, we have the expression of the volume form on $X_\zeta^*$ as:

$$vol_{X_\zeta^*} \quad = \quad (\mu_\zeta)^* \Phi_\zeta A \wedge d(\mu_\zeta)_1 \wedge d(\mu_\zeta)_2 \wedge d(\mu_\zeta)_3.$$

On the other hand, thanks to Proposition 5.5, there exist two constants $C_1, C_2 \in \mathbb{R}$ such that

$$\|R^E\| \quad \leq \quad C_2 \|q_0\|^{-5} \quad on \quad X_\zeta \backslash K_\zeta$$

where we define $K_\zeta := \{ [(q_n)] \in X_\zeta \mid \|q_0\| \leq C_1 \}$. We see that there exists a compact set $K \subset Im\mathbb{H}$ such that $K_\zeta \subset \mu_\zeta^{-1}(K)$. Since the hyper-Kähler moment map $\mu_\zeta$ is proper (Theorem4-8 in [6]), $\mu_\zeta^{-1}(K)$ turns out to be compact in $X_\zeta$. Let us define $K'_\zeta := \mu_\zeta^{-1}(K)$, and then it suffices to see that the following value

$$\int_{X_\zeta^* \backslash K'_\zeta} \|R^E\|^2 vol_{X_\zeta^*} \quad \leq \quad C_2^2 \int_{X_\zeta^* \backslash K'_\zeta} \|q_0\|^{-10} (\mu_\zeta)^* \Phi_\zeta A \wedge d(\mu_\zeta)_1 \wedge d(\mu_\zeta)_2 \wedge d(\mu_\zeta)_3 \quad \leq \quad C_3 \int_{Im\mathbb{H} \backslash K} \frac{1}{r^5} \Phi_\zeta vol_{Im\mathbb{H}}$$

is finite, where $r$ is the radius function on $Im\mathbb{H}$ and we use the fact that $\|q_0\|^{-10} (\mu_\zeta)^* \Phi_\zeta$ is $S^1$-invariant.

Let us take a sufficiently small $\epsilon > 0$ which satisfies the following conditions: When we put $B_n := B(\lambda(\zeta)_n; \epsilon)$,

$$(1) B_n \cap B_m = \emptyset \text{ for } n \neq m, \quad (2)\ B_0 = B(0;\epsilon) \subset K, \quad (3) B_n \subset Im\mathbb{H} \backslash K \text{ for } n \in S.$$

where we define $S := \{ n \in \mathbb{Z} \mid \lambda(\zeta)_n \in Im\mathbb{H} \backslash K \}$. Note that such a $\epsilon$ actually exists because $\|\lambda(\zeta)_n\| = O(n^2)$, and also note that $0 \notin S$ by the condition (2).

For any fixed $n \in S$, it follows that

$$\begin{aligned}
\int_{B_n} \frac{1}{r^5} \Phi_\zeta vol_{Im\mathbb{H}} \quad &= \quad \int_{B_n} \frac{1}{r^5} \frac{1}{\|\eta - \lambda(\zeta)_n\|} vol_{Im\mathbb{H}} + \int_{B_n} \frac{1}{r^5} \sum_{m \neq n} \frac{1}{\|\eta - \lambda(\zeta)_m\|} vol_{Im\mathbb{H}} \\
&\leq \quad \frac{1}{(\|\lambda(\zeta)_n\| - \epsilon)^5} \int_{B(0;\epsilon)} \frac{1}{r} vol_{Im\mathbb{H}} + \frac{1}{(\|\lambda(\zeta)_n\| - \epsilon)^5} \int_{B_n} \sum_{m \neq n} \frac{1}{\|\eta - \lambda(\zeta)_m\|} vol_{Im\mathbb{H}} \\
&= \quad \frac{C_4}{(\|\lambda(\zeta)_n\| - \epsilon)^5} + \frac{1}{(\|\lambda(\zeta)_n\| - \epsilon)^5} \frac{4\pi\epsilon^3}{3} \sum_{m \neq n} \frac{1}{\|\lambda(\zeta)_n - \lambda(\zeta)_m\|}.
\end{aligned}$$

Here we use the mean value property of harmonic functions at the second equality.

**Lemma 7.1.** *There exists a constant $C_5$ such that, for any $n \in S$,*

$$\sum_{m \neq n} \frac{1}{\|\lambda(\zeta)_n - \lambda(\zeta)_m\|} \quad \leq \quad C_5.$$

*Proof.* For the sake of simplicity, we only prove the case with $\zeta = 0$. In this case, we have

$$\lambda(\zeta)_n = \begin{cases} in^2 & (n \geq 0) \\ -in^2 & (n < 0) \end{cases} .$$

Let us decompose $S$ as $S = S^+ \cup S^-$ where we define $S^+ := \mathbb{Z}_{>0} \cap S$ and $S^- := \mathbb{Z}_{<0} \cap S$. For $\forall n \in S^+$, we have

$$\begin{aligned}
\sum_{n \neq m} \frac{1}{\|\lambda(\zeta)_n - \lambda(\zeta)_m\|} &= \left( \sum_{m<0} + \sum_{0 \leq m < n} + \sum_{n<m} \right) \frac{1}{\|\lambda(\zeta)_n - \lambda(\zeta)_m\|} \\
&\leq \sum_{m<0} \frac{1}{n^2+m^2} + \int_0^{n-1} \frac{dx}{n^2-x^2} + \frac{1}{n^2-(n-1)^2} + \int_{n+1}^{\infty} \frac{1}{x^2-n^2} + \frac{1}{(n+1)^2-n^2} \\
&\leq \zeta(2) - \left[ \frac{1}{2n} \log \left| \frac{x-n}{x+n} \right| \right]_{x=0}^{x=n-1} + \frac{1}{2n-1} + \left[ \frac{1}{2n} \log \left| \frac{x-n}{x+n} \right| \right]_{x=n+1}^{x=\infty} + \frac{1}{2n+1} \\
&= \zeta(2) + \frac{1}{2n} \log(2n-1) + 1 + \frac{1}{2n} \log(2n+1) + 1 \\
&\leq C_5,
\end{aligned}$$

where $\zeta(2)$ is the value of Riemann $\zeta$-function at 2. The same estimate can be applied for $\forall n \in S^-$. □

Thus, we have

$$\sum_{n \in S} \sum_{m \neq n} \frac{1}{(\|\lambda(\zeta)_n\| - \epsilon)^5} \frac{1}{\|\lambda(\zeta)_n - \lambda(\zeta)_m\|} \leq \sum_{n \in S} \frac{C_5}{(\|\lambda(\zeta)_n\| - \epsilon)^5} < \infty.$$

and it follows that

$$\int_{\cup_{n \in S} B_n} \frac{1}{r^5} \Phi_\zeta vol_{Im\mathbb{H}} = \sum_{n \in S} \int_{B_n} \frac{1}{r^5} \Phi_\zeta vol_{Im\mathbb{H}} < \infty.$$

**Lemma 7.2.** *There exists a constant $C_6$ such that*

$$\Phi_\zeta(\eta) \leq C_6, \qquad \forall \eta \in Im\mathbb{H} \backslash (\cup_{n \in S} B_n).$$

*Proof.* For simplicity, we may assume $\zeta = 0$ as in the previous lemma.
For any $l \in \mathbb{N}$, we define $D^l := \{\eta \in Im\mathbb{H} \backslash (\cup_{n \in \mathbb{Z}} B_n) \mid l^2 \leq \|\eta\| \leq (l+1)^2\}$. Then, for any $\eta \in D^l$, we have

$$\begin{aligned}
\sum_{m \in \mathbb{Z}} \frac{1}{\|\eta - \lambda(\zeta)_m\|} &= \left( \sum_{|m|=l,l+1} + \sum_{|m|<l} + \sum_{|m|>l+1} \right) \frac{1}{\|\eta - \lambda(\zeta)_m\|} \\
&\leq \frac{4}{\epsilon} + 2 \sum_{0 \leq m < l} \frac{1}{\|\eta - im^2\|} + 2 \sum_{m > l+1} \frac{1}{m^2 - (l+1)^2} \\
&\leq \frac{4}{\epsilon} + 2 \sum_{0 < m \leq l} \frac{1}{l^2 - (l-m)^2} + 2 \sum_{m > l+1} \frac{1}{(m-(l+1))^2} \\
&= \frac{4}{\epsilon} + 2 \sum_{0 < m \leq l} \frac{1}{m(2l-m)} + 2\zeta(2) \\
&\leq \frac{4}{\epsilon} + 2 \sum_{0 < m \leq l} \frac{1}{m^2} + 2\zeta(2) \\
&\leq \frac{4}{\epsilon} + 2\zeta(2) + 2\zeta(2) =: C_6.
\end{aligned}$$

□

Thanks to the above lemma, if we put $U := Im\mathbb{H} \backslash (\cup_{l \in S} B_l \cup K)$, we have

$$\int_U \frac{1}{r^5} \Phi_\zeta(\eta) vol_{Im\mathbb{H}} \leq C_6 \int_U \frac{1}{r^5} vol_{Im\mathbb{H}} < \infty.$$

Summarizing above, finally we obtain

$$\int_{Im\mathbb{H}\backslash K} \frac{1}{r^5}\Phi_\zeta(\eta) \quad < \quad \infty,$$

and therefore we complete the proof of the following theorem:

**Theorem 7.3.** *Let $E$ be an instanton constructed from a finite dimensional ADHM data $(V,W,B,\Psi)$. Then, the $L^2$-norm of the curvature $R^E$ of the induced connection on $E$ is finite.*

**Corollary 7.4.** *Under the same assumption in Theorem 7.3, $L^p$-norm of the curvature $R^E$ is finite for any $p > \frac{6}{5}$.*

*Proof of corollary 7.4.* Due to the above calculations, we have

$$\begin{aligned}
\int_{X_\zeta} \|R^E\|^p vol_{X_\zeta} \ < \ \infty \quad &\Leftarrow \quad \int_{Im\mathbb{H}\backslash K} r^{-\frac{5}{2}p} vol_{Im\mathbb{H}} \ < \ \infty, \ \sum_{n\in\mathbb{Z}} \frac{1}{(n^2)^{5p/2}} \ < \ \infty \\
&\Leftrightarrow \quad -\frac{5}{2}p+2 < -1, \ \ 5p > 1 \\
&\Leftrightarrow \quad p > \frac{6}{5}.
\end{aligned}$$

□

# 8 Examples

In this section, we give some examples of intantons constructed from finite dimensional data by using the result of section 3, and we see that analytic sheaves can be constructed from infinite dimensional data which is associated with the points on $\mu^{-1}(\zeta)$ by using the result in section 4. Note that we only consider the case where $\zeta = 0$.

To begin with, we consider finite dimensional ADHM data. The construction shows that for any finite dimensional ADHM data $(V, W, \mathcal{B}, \Psi)$, we have

$$c_1(E) \quad = \quad c_1\left((V \otimes \mathcal{L} \otimes Q)^{S^1}\right) + c_1\left((W \otimes \mathcal{L})^{S^1}\right) - 2c_1\left((V \otimes \mathcal{L})^{S^1}\right).$$

For the sake of convenience, we decompose $V = \oplus_{i \in S_V} \mathbb{C}^{v_i} \otimes V_i$ and $W = \oplus_{i \in S_W} \mathbb{C}^{w_i} \otimes V_i$. Then it follows that

$$\begin{aligned} c_1(E) &= \sum_{i \in S_V} v_i(c_1(\mathcal{L}_{-i+1}) + c_1(\mathcal{L}_{-i-1})) + \sum_{i \in S_W} w_i c_1(\mathcal{L}_{-i}) - 2\sum_{i \in S_V} v_i c_1(\mathcal{L}_{-i}) \\ &= \sum_i (w_i - 2v_i + v_{i+1} + v_{i-1}) c_1(\mathcal{L}_{-i}) \end{aligned}$$

where we put $v_i = 0$ for $i \notin S_V$ and $w_j = 0$ for $j \notin S_W$.

**Example 8.1.** *For $V = 0$ and $W = V_{-n}$, we have an instanton line bundle $E = \mathcal{L}_n$. (In this case there is no other choice than $\mathcal{B} = 0$ and $\Psi = 0$, and the ADHM equations are empty.)*

**Example 8.2.** *Consider the case with $V = V_n$ and $W = V_n$. In this case we have to find $\Psi_1, \Psi_2 : V_n \to V_n$ which satisfies the ADHM equations. By regarding $\Psi_1, \Psi_2 \in \mathbb{C}$, the ADHM equations are equivalent to*

$$\begin{aligned} \Psi_1\Psi_2 &= -((\zeta_V)_{\mathbb{C}})_n = 0, \\ |\Psi_1|^2 - |\Psi_2|^2 &= -((\zeta_V)_{\mathbb{R}})_n = \begin{cases} 2n+1 & (n \geq 0) \\ -2n+1 & (n < 0) \end{cases} = 2|n| + 1. \end{aligned}$$

*Clearly there actually exist solutions for the ADHM equations. In this case* $\dim E = \dim W = 1$ *and $c_1(E) = -c_1(\mathcal{L}_{-n}) = c_1(\mathcal{L}_n)$, and therefore we have $E \cong \mathcal{L}_n$.*

Next we give examples of infinite dimensional ADHM data associated withe the points on $\mu^{-1}(\zeta)$.

**Example 8.3.** *Let us consider the case in which $V = L^2(S^1)$ and $W = V_0$, and take any linear map $\Psi_1 \in Hom(V_0)$ and any point $\xi \in \mu^{-1}(0)$. Then we have an infinite dimensional ADHM data $(L^2(S^1), V_0, \xi, (\Psi_1, 0))$. Note that, since injectivity of $\mathcal{D}$ is not guaranteed in this case, $E = Ker\tau/Im\sigma$ may not be a vector bundle but at least an analytic free sheaf over $X_\zeta$. In addition, the rank of its stalk at $p \in X_\zeta$ is given by* $\mathrm{rank}(E_p) = 1 + \mathrm{rank}(Ker(\sigma)_p) + \mathrm{rank}(Ker(\tau^*)_p) < \infty$ *by Proposition 4.5.*

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